

AI Learning Companion

Master Specification v3.0

Working persona options: “Talk to Jana” or “Talk to Sami”

Product ambition

Build a profession-aware, exam-aware, bilingual AI learning operating system that acts as tutor, study planner, navigator, assessor, coach, support agent and long-term learning memory. Launch with OET, then expand to IELTS, PTE, TOEFL and B2B/white-label learning.

Pillar	Non-negotiable outcome
PERSONALISE	Every learner receives an adaptive plan, memory, error profile and next-best action.
TEACH	The AI is grounded in approved Rule Books, sessions, workshops, corrections and all eligible materials.
ACT	It opens resources, starts practice, updates plans, saves notes and guides users inside the product.
MONETISE	Free demonstrates value; paid tiers unlock the learning engine. Heavy AI actions use controlled allowances/top-ups.
SCALE	The same core supports multiple exams, professions, languages, institutions and white-label deployments.

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1. Executive Summary

One sentence definition

This is not a chatbot page. It is a persistent AI mentor embedded across the website, web app and native apps, with the same identity, memory and permissions everywhere.

- Launch focus: OET, with full profession-specific intelligence and knowledge of the complete approved OET learning ecosystem.
- Future scope: IELTS, PTE, TOEFL and other exam/curriculum packs without rebuilding the product core.
- Primary commercial model: independent AI subscription tiers layered on top of content/course entitlements.
- Primary learner promise: the AI always knows who the candidate is, what they need, what they have already done, where their weaknesses are, and what the best next action is.
- Primary business promise: every expensive AI behavior is metered, routed and capped so paid plans target strong gross margins instead of uncontrolled API spending.

Core product architecture

Layer	Required capability
Candidate Intelligence	Profile, exam date, target, profession, scores, habits, preferences, history, confidence and availability.
Knowledge Brain	Official exam knowledge + Dr Hesham methodology + complete eligible course materials + platform/support knowledge.
Learning Engine	Study plan, next-best action, mastery map, error DNA, adaptive drills, spaced reinforcement and readiness.
Tutor Engine	Writing, Speaking, Reading, Listening, grammar, vocabulary, voice, multimodal and mock support.
Action Layer	Deep links, start practice, open resource, save notes, update plan, reminders, tutor/support handoff.
Commercial Layer	Entitlements, AI tiers, usage allowances, premium actions, top-ups, upgrade cards, revenue/cost dashboards.
Trust Layer	Source hierarchy, version awareness, anti-hallucination, privacy, tenant isolation, access control and auditability.

Version 3.0 consolidation - what changed

Area	v3.0 decision
Economics	Pricing is now evaluated after VAT, app-store/payment fees and AI variable cost; voice is no longer a large unmetered base allowance.
AI Credits	The existing AI credit system becomes the single user-facing metering currency. No second “Power Actions” currency is introduced.
Security	Adds proprietary-content exfiltration defence, prompt-injection resistance, anomaly detection and red-team requirements.
Privacy & legal	Adds a real privacy/compliance plan, voice consent, sub-processor/DPA requirements, retention rules, trademark/app-store review gates.
Technical delivery	Adds retrieval architecture, model routing, caching, memory compaction, vendor abstraction, observability and abuse controls.
Quality	Adds profession/subtest golden sets, regression testing, grading calibration, hallucination measurement and Arabic-specific evaluation.
Content operations	Adds a complete ingestion/approval pipeline for Rule Books, videos, transcripts, PDFs, recalls and future updates.
Scope	The full vision remains preserved, but Stage 1 is deliberately reduced to a monetisable OET core; later features are explicitly phased, not silently dropped.
Final verdict	Broad feature ideation is complete enough to proceed. Further work is validation/sign-off, not another brainstorming round.
v3.0 final controls	Defines cost-per-credit economics and top-up floors; adds UK subscription-contract readiness, core data schemas, owned-audience launch economics, scope-versus-budget gating, clinical/distress boundaries, WCAG 2.2 AA, credit refund/reversal controls, emergency kill switches, a defined live beta, and uptime/incident operations.

The full v1.0/v2.0 feature inventory is preserved. Version 3.0 closes the remaining economic, compliance, data-model, owned-audience, budget-control, healthcare-boundary and operational-control gaps without expanding the product feature scope.

2. Product Principles and Non-Negotiables

Principle	Requirement
AI everywhere, not one AI page	The assistant must be available as a floating chat, full-screen tutor, side panel, in-video helper, writing copilot, speaking room, results coach, dashboard coach and support assistant.
Personalisation before generic answers	Every plan, explanation and recommendation should use the candidate profile when relevant.
Profession-first OET logic	Profession is a core dimension of the knowledge model, content access, examples, role plays and writing/speaking guidance.
Source-grounded teaching	Approved Rule Books and course methodology override generic internet-style advice for teaching strategy.
Official facts remain official	Booking, scoring, exam format, regulator requirements and other current facts must come from verified current sources and be separated from teaching strategy.
Entitlement before retrieval	The AI must never leak paid course content to users who do not own that content.
Action over explanation	When the product can perform a useful action, the AI should offer the action, not only describe where to click.
Free proves value; paid creates habit	Free is a conversion experience, not a complete substitute for paid AI.
Cost-aware by design	Simple navigation must not consume the same expensive compute as deep assessment or voice simulation.
One identity everywhere	The learner should continue the same conversation and learning journey across website, web app, iOS, Android, Windows and macOS.

3. Working Brand and Persona Direction

The current preferred human-style persona options are:

Option	Primary CTA	Strength	Brand note
Jana	Talk to Jana	Warm, human, easy across Arabic and English; excellent for voice and coaching.	Use as persona pending trademark/domain clearance.
Sami	Talk to Sami	Friendly, simple, familiar across the Middle East and internationally.	Use as persona pending trademark/domain clearance.

Recommended naming architecture

Treat Jana/Sami as the user-facing AI persona. The legal/master product brand may remain separate if trademark ownership, B2B licensing or valuation requires a more distinctive mark. The UI can still use “Talk to Jana” or “Talk to Sami” everywhere.

- Never describe the product publicly as only an “AI chatbot”. Preferred language: AI Learning Companion, AI Tutor, AI Mentor, Personal Learning Intelligence or AI Learning OS.
- Ultimate positioning: “Your Personal AI Mentor” - the tier that actively manages the learner’s preparation, not just answers questions.

4. Users, Roles and Product Surfaces

User roles

Role	Capabilities
Anonymous visitor	Limited demo; lead capture; no persistent academic profile until registration.
Registered Free learner	Basic profile and capped free AI.
Paid learner	AI tier + content entitlements determine capabilities.
Tutor	Candidate summaries, handoff, tutor notes, teaching overrides where authorised.
Admin	Content, pricing, entitlements, AI quality, support, analytics and revenue controls.
Super Admin / B2B owner	Institution/tenant control, branding, policies, knowledge packs, seats, budgets and audit logs.

Required surfaces

- Public website floating assistant and landing-page demo.
- Web app floating assistant + full-screen AI Tutor.
- Context side panel beside Reading/Listening questions.
- AI inside Writing editor.
- Voice-first AI inside Speaking practice and assessment.
- AI beside videos with current-video and timestamp awareness.
- AI on results/analytics pages.
- AI on learner dashboard and “What should I do next?” card.
- Android and iOS apps.
- Windows and macOS apps or equivalent desktop experience.

- Optional future channels: email/push summaries, WhatsApp/Telegram integrations, LMS/enterprise integrations - all using the same identity and entitlements.

5. Candidate Intelligence Profile and Onboarding

Goal

Within the first useful interaction, the assistant should know enough to personalise. It should progressively enrich the profile rather than forcing a long form before the learner sees value.

Profile area	Examples of stored learning context
Identity & access	Account, AI tier, purchased packages, profession, active exam, language preference.
Goal	Target score/grade, regulator/university/country goal, exam date, first attempt vs resit.
Baseline	Previous official results, mock averages, self-rated confidence and diagnostic performance.
Availability	Study minutes/day, work shifts, days off, preferred study times, planned travel/leave.
Skill profile	Reading/Listening/Writing/Speaking subskills, grammar, vocabulary, timing, pronunciation, confidence.
Learning preference	Arabic/English/mixed, concise/detailed, examples vs rules, voice vs text, preferred practice style.
History	Completed lessons, watched videos, mocks, submitted writing, speaking attempts, tutor feedback, missed tasks.
Behavior	Answer-changing pattern, confidence calibration, speed, consistency, streaks and follow-through.
Readiness	Current mastery, trend, blockers, latest recommended action and risk flags.

- Allow manual editing of exam date, profession, target and availability.
- Allow candidate to view, edit, delete and reset AI memory.
- Support multiple exam goals over time, with one active preparation journey and preserved historical journeys.
- Extract scores from uploaded screenshots/reports after user confirmation.
- If exam date changes, automatically rebuild the plan and explain the impact.

6. OET Knowledge Brain: Everything the AI Must Study

No arbitrary omissions

Every approved educational asset that can help an entitled learner must be ingestible, searchable, source-tagged, versioned and connected to the relevant profession, subtest, skill and package.

Knowledge source	Required coverage
Dr Hesham Rule Books	All approved Writing and Speaking Rule Books by profession; future Reading/Listening/grammar/vocabulary rule sets; corrections and exceptions.
Recorded teaching	All relevant teaching sessions, course videos, workshops, live-session recordings, correction sessions and their transcripts/timestamps.
Course materials	PDFs, notes, slides, handouts, exercises, question banks, model answers, explanations, Basic English support material and profession-specific content.
Tutor Book & dictionaries	Tutor Book content, reading dictionaries, common-word lists, vocabulary sets, recall-linked study material and approved explanations.
Recalls & practice banks	Eligible Listening/Reading/Writing/Speaking recall content, mock/practice series, associated explanations and resource metadata.
Platform map	Every page, menu, course, tab, resource, video, exam, subscription state, credit state and deep-link destination.
Support knowledge	Registration, OTP, devices, access, expiry, package rules, credits, troubleshooting, billing/support FAQs and escalation pathways.
Official exam knowledge	Current official format, scoring, booking, ID, policies, delivery mode and other factual exam information, versioned by effective date.
Candidate-specific knowledge	Scores, errors, attempts, notes, tutor feedback, plan, vocabulary, memory, confidence and behavior.
Admin-approved updates	Teacher overrides, newly approved rules, content changes, corrections and policy updates.

Knowledge precedence rules

1. Official exam facts/policies take precedence for factual exam requirements.
2. Dr Hesham approved Rule Books and teaching methodology take precedence for teaching strategy.

3. Profession-specific rules take precedence over generic OET rules when both apply.
4. Newer approved versions take precedence over obsolete content; old versions remain auditable.
5. Candidate-specific tutor instructions may customise the study plan but must not rewrite official exam facts.
6. If sources conflict, the AI should explicitly state the conflict and use the highest-authority/current source rather than inventing a compromise.

7. Content Intelligence, Search and Video Awareness

- Universal semantic search across Rule Books, transcripts, videos, PDFs, notes, workshops, corrections, vocabulary and FAQs.
- Exact source recall: “Where did Dr Hesham explain this?” returns the relevant session and, when available, exact timestamp.
- In-video assistant understands the current video and playback timestamp.
- “Summarise the last 10 minutes”, “quiz me on what I just watched”, “save this rule”, and “show me the example from this lesson”.
- Automatic session notes: rules, warnings, examples, vocabulary, mini-quiz and candidate-specific takeaways.
- Content-gap detection: identify frequent questions that lack a dedicated lesson or clear resource.
- Learning-gap detection: identify rules candidates still miss even after viewing the relevant content.
- New-content pipeline: transcript, chaptering, rule extraction, FAQs, examples, tags and quizzes are proposed automatically and activated only after approval.
- Source citations/labels in educational answers where useful; paid source details must not expose locked content to unauthorised users.

8. Adaptive Study Plan and Next-Best-Action Engine

Primary killer feature

The assistant should be able to answer “I have 25 minutes now - what should I do?” with the single most valuable action for that candidate, then open the correct task.

- Initial diagnostic plan
- Daily plan
- Weekly plan
- 90/60/30-day plans
- 14-day intensive plan
- 7-day emergency plan
- 3-day final revision
- Exam-eve plan
- Post-result/resit plan
- Single-subtest recovery plan

- Working-doctor/shift-worker plan
- Weekend-heavy plan
- 20-min/day plan
- 2-hour/day plan
- Missed-days recovery plan
- Travel/holiday-adjusted plan

Plan inputs and behavior

- Use exam date, target, profession, latest scores, trend, completion, recurring errors, study time, current confidence and content entitlements.
- Replan automatically after new results, missed tasks, changed work schedule, changed exam date or clear mastery improvement.
- Do not simply carry overdue tasks forever; reprioritise and explain what was dropped or delayed.
- Prefer high-impact weaknesses over mechanically finishing every course item in order.
- Generate the exact next resource or exercise and offer a one-click deep link.
- Change behavior near the exam: shift from content acquisition to rehearsal, error review, confidence and logistics.
- Ultimate tier should support continuous next-best-action decisions and proactive plan updates.

9. Memory, Error DNA and Learning Fingerprint

Three memory layers

Layer	Examples	User control
Conversation memory	Current task, recent questions, temporary context.	Automatic + clear/reset chat.
Learning memory	Scores, errors, mastery, plan history, vocabulary, tutor feedback, completed work.	View/edit/delete/reset learning profile.
Journey memory	Longitudinal pattern: score changes, interventions, what worked, recurring blockers.	Long-term on higher tiers; export/delete available.

Error DNA

- Track recurring grammar, writing, speaking, reading, listening, timing, vocabulary and reasoning errors.
- Quantify which error categories are costing the most marks over recent attempts.
- Provide “Train me on my mistakes” sessions generated from the candidate’s own history.
- Re-test old errors using spaced reinforcement until mastery is demonstrated.
- Separate knowledge gaps from speed problems, confidence problems, attention problems and exam-strategy problems.

Learning Fingerprint

- Preferred language/mix and explanation depth.
- Response to examples vs abstract rules.
- Voice vs text learning preference.
- Typical forgetting interval and best review spacing.
- Confidence vs accuracy calibration.
- Performance under time pressure vs untimed practice.
- Whether the learner changes correct answers, rushes, overthinks or repeatedly misses distractors.
- Which interventions historically improved the learner most.

10. Teaching Modes and Core Conversational Behaviors

Mode	Behavior
Quick Answer	Short, direct response with optional deeper explanation.
Detailed Tutor	Step-by-step teaching with examples.
Socratic Tutor	Hints and questions; learner discovers answer rather than receiving it immediately.
Examiner	Strict evaluation, no hints during active mock.
Coach	Progress, accountability, prioritisation and plan adjustment.
Dr Hesham Method	Uses approved course methodology and cites the relevant rule/source where useful.
Arabic Explanation	Arabic explanation while preserving English medical/exam terminology when requested.
English-only	Immersion and exam-style English.
Practice Generator	Creates targeted questions/drills with adaptive difficulty.
Revision	Fast recap of the learner's own weak rules and saved notes.

- Quick actions after explanations: Explain simpler, Explain in Arabic, Give another example, Quiz me, Save this, Add to plan, Open lesson, Ask tutor.
- Support natural Arabic-English code-switching without forcing translation.
- Remember the learner's preferred style but allow per-message overrides.

11. Writing AI - Full Tutor and Assessment System

- Profession-specific task interpretation and recipient awareness.
- Purpose analysis and purpose-paragraph coaching.

- Case-note selection: relevant vs irrelevant information and rationale.
- Organisation, chronology, paragraphing and information hierarchy.
- Conciseness and removal of unnecessary details.
- Grammar, articles, prepositions, tense choice, agreement, punctuation and sentence structure.
- Clinical-to-lay language adaptation and appropriate professional register.
- Vocabulary, collocations, clarity and naturalness.
- OET-style criteria-based evaluation and estimated performance band/score where appropriate.
- Sentence-level corrections with “candidate version → improved version → why”.
- Guided writing mode: paragraph-by-paragraph construction.
- Hint mode: identify what must be fixed without rewriting it for the candidate.
- Rewrite mode: improve selected text while explaining the rule.
- Compare mode: compare two versions and explain which is stronger.
- Strict examiner mode vs teaching mode.
- Recurring-error tracking and mark-loss attribution.
- Personalised micro-lessons generated from the candidate’s repeated writing errors.
- Ability to connect feedback to the exact Rule Book rule/session/timestamp.
- Human tutor escalation when uncertainty, complex judgment or package entitlement requires it.
- Post-correction action plan: the next 1-3 writing drills rather than feedback only.

12. Speaking AI - Voice, Role Play and Assessment

- Full voice conversation with low-friction start/stop and transcript.
- Profession-specific patient/relative/carer/colleague role cards and realistic scenarios.
- Patient personalities: anxious, angry, reluctant, confused, talkative, quiet, worried parent, demanding relative and mixed-complexity cases.
- Practice modes: friendly, standard, difficult patient, strict exam, surprise role card, full mock, fluency, pronunciation, empathy, information gathering and treatment explanation.
- Assess completion of role-card tasks, information gathering, explanations, empathy, structure, fluency, grammar, vocabulary and communication effectiveness.
- Pronunciation feedback: intelligibility, stress, pace, fillers, pauses and repeated mispronunciations.
- Post-session replay: transcript, missed tasks, better alternatives and targeted next practice.
- Allow “pause role play and coach me” in practice mode; disable coaching in exam mode.
- Generate new role plays based on the candidate’s profession and recurring weaknesses.
- Track repeated communication problems across sessions and add them to Error DNA.
- Voice time should be metered separately from cheap text interactions for profitability.

13. Reading and Listening AI Coaches

Reading

- Analyse Part A/B/C separately and by question type/subskill.
- Track time per question/part, evidence selection, distractor susceptibility and answer changes.
- Ask confidence after selected answers and analyse confidence-vs-accuracy calibration.

- Explain why the selected answer is wrong and why alternatives are wrong, not only reveal the correct answer.
- Generate targeted inference, scanning, paraphrase, vocabulary and timing drills.
- Recommend the exact next reading practice based on the weakness profile.
- Compare performance across recent tests and explain why the score changed.

Listening

- Analyse Parts A/B/C separately.
- Track spelling, missed words, distractors, accent difficulty, prediction failures, concentration, paraphrase and synonym problems.
- Generate targeted dictation, medication spelling, number/date, accent and paraphrase drills.
- Create personal listening vocabulary from missed words.
- Use candidate mistakes for spaced re-testing.
- Explain score changes using evidence from attempts, not generic advice.
- Recommend relevant approved listening content/podcasts/material from the candidate's plan and entitlement.

14. Grammar, Vocabulary and Micro-Learning

- Build “My Grammar Weaknesses” instead of forcing every learner through a generic grammar course.
- Generate rule explanations, profession-relevant examples, mini exercises and adaptive re-tests.
- Build “My Vocabulary Brain” from reading/listening errors, writing/speaking output, dictionaries, recalls and saved words.
- Vocabulary cards: meaning, pronunciation, collocation, synonym, paraphrase, clinical vs lay usage and example sentence.
- Spaced repetition scheduler based on actual errors and forgetting patterns.
- Micro-lessons of 2-5 minutes for a single rule or weakness, followed immediately by a short check.
- Allow “make it harder/easier”, “make it Nursing-specific”, “use my mistakes” and “test me without hints”.

15. Mock, Exam Mode and Test-Day Intelligence

- Full computer-based exam rehearsal with realistic timing and navigation.
- Exam mode disables hints, answer explanations, transcripts and coaching until submission.
- Practice mode may provide hints according to the learner's selected teaching mode.
- After submission, provide section-level analytics, error causes, timing analysis and next actions.
- Test-day rehearsal mode: full simulation plus logistics checklist.
- Final 24-hours mode: revision, logistics, warm-up and confidence; no unnecessary new content overload.
- Current exam-format/version awareness based on exam date.
- Result-day assistant: ingest results, celebrate success appropriately, or build a resit recovery plan.

- Academic-integrity guardrails: never act as a live answer engine during a real official examination.

16. Multimodal and File Intelligence

- Understand screenshots, score reports, question images, PDFs, handwritten notes, writing letters, role cards and approved audio recordings.
- Extract exam date/score data from uploaded evidence and ask for confirmation before saving it to profile.
- Explain an uploaded question or document in the learner's preferred language.
- Compare versions of a writing submission.
- Turn uploaded notes into study cards or a revision checklist.
- Detect likely sensitive/patient-identifying information in uploads and warn the learner before storing or sharing it.
- File/image usage is tiered and metered because large files can create high variable cost.

17. Platform Navigator and Action-Taking AI

Required behavior

A learner asking “Where is X?” should receive a direct Open/Start button whenever the platform can resolve the destination. The AI should not force the learner to manually follow a long path unless necessary.

- Open exact course/module/video/PDF/exam/question.
- Open a video at a relevant timestamp.
- Start recommended practice.
- Continue last activity.
- Save a rule to personal notes.
- Add a word to vocabulary.
- Add an activity to today's plan.
- Mark a plan item complete where the product has evidence.
- Set a study reminder or exam countdown notification.
- Open upgrade/checkout card.
- Create a support request with context.
- Prepare a tutor handoff summary.
- Open workshop/booking page.
- Report a wrong answer/content issue.
- Show remaining AI allowance/credits and explain what consumes them.

18. Context-Aware AI Inside Every Screen

- Know the current page, active exam, active question, selected answer, visible content and current course where privacy/security permits.

- “Explain this” should refer to the current item without asking the learner to copy-paste it again.
- Results page: explain score pattern and next action.
- Course page: answer from the active module and relevant Rule Book.
- Video page: answer from current video/timestamp.
- Writing editor: answer about the selected sentence/paragraph or overall letter.
- Speaking room: remain in role-play context until the learner exits.
- Never let screen context bypass content entitlement rules.

19. Readiness, Mastery and Personal Analytics

- Overall readiness score with subtest readiness and transparent drivers.
- Mastery map by skill/subskill using green/amber/red or equivalent states.
- Trend lines for score, accuracy, timing and confidence.
- “What is stopping me from being exam-ready?” diagnosis.
- “Why did my score change?” evidence-based analysis across recent attempts.
- Personal bests and milestone recognition.
- Anonymous cohort benchmarking only when statistically responsible and privacy-safe.
- Avoid presenting readiness as a guaranteed pass probability; show uncertainty when appropriate.
- Weekly progress report summarising wins, risks, next priorities and planned actions.

20. Proactive Coaching, Calendar and Notifications

- Inactivity nudges when preparation is at risk.
- Exam countdown messages with plan adaptation.
- Reminder when a key subtest has not been practised recently.
- New relevant content/recall notifications according to entitlement and preferences.
- Calendar-aware scheduling of study blocks, workshops, private sessions and exam date.
- If the learner reports a long work shift or day off, adjust the plan immediately.
- Weekly progress summary by app/email/push according to notification consent.
- Ultimate: proactive next-best-action coaching and continuous replanning; lower tiers receive less frequent/adaptive support.
- Avoid spam: notification frequency and quiet hours are user-controlled.

21. Human Tutor Collaboration, Admin AI and Content Operations

Tutor collaboration

- Pre-session tutor brief: last scores, trend, recurring errors, plan adherence and recommended session focus.
- Post-session tutor note becomes part of learner memory and future plan.
- One-click candidate summary for handoff without forcing the learner to repeat their history.

- Escalation when the AI is uncertain, the user requests a human, or a premium human service is required.

Admin AI

- Ask aggregate questions such as “What are Nursing candidates struggling with this week?”
- AI quality dashboard: incorrect answers, low-confidence answers, thumbs-down, unresolved questions, source conflicts and hallucination flags.
- Content-gap dashboard: high-volume questions with weak/no content coverage.
- Teaching-gap dashboard: concepts still frequently missed after lesson completion.
- Usage/cost/revenue dashboard by tier, feature, profession and cohort.
- Admin can correct a rule once through an authoritative override and have the change propagate after approval.
- Maintain version history, effective date, approver and rollback for teaching rules and critical knowledge.

22. Trust, Accuracy, Safety and Privacy Requirements

Area	Requirement
Anti-hallucination	If verified information is insufficient, say so. Never fabricate an official rule, course rule, score requirement, page location or entitlement.
Source transparency	Differentiate official exam facts from Dr Hesham teaching strategy. Surface source labels/citations when they add trust.
Version awareness	Exam knowledge is date/version aware. The learner's test date determines which version applies.
Entitlement security	All retrieval and generated answers respect purchased content, profession and package access.
Privacy	User-visible memory controls, deletion/export, data minimisation and appropriate retention settings.
Tenant isolation	Future B2B institutions cannot access another institution's knowledge or learner data.
Sensitive uploads	Warn/redact when likely real patient-identifying information is uploaded; educational use is not medical advice.
Integrity	AI help is disabled or constrained during protected mocks/real exams where assistance would be inappropriate.
Auditability	Critical admin changes, entitlement decisions, rule updates and AI quality events are logged.

Area	Requirement
Accessibility	Target WCAG 2.2 AA across all learner-facing website, web-app and native-app surfaces, including captions, transcripts, keyboard navigation, scalable text, contrast, focus states, speech-to-text/text-to-speech alternatives and accessible error/paywall flows.
Content exfiltration defence	Never reproduce proprietary Rule Books or paid materials in bulk, verbatim or by multi-turn reconstruction. Enforce per-source verbatim-span limits, per-user retrieval-volume limits, anomaly detection and refusal patterns for extraction attempts.
Prompt-injection defence	Treat retrieved documents, uploads and user text as untrusted data. Protect system prompts, hidden policies, credentials and internal routing instructions from extraction or override attempts.
Canary / leak detection	Permit approved watermark/canary strings in protected source material and monitor for suspicious verbatim leakage. Any canary hit becomes a security event.
Vendor data terms	Before production, document each model/voice/hosting vendor's retention and training terms. Proprietary teaching content must not be used for vendor training unless explicitly approved in writing.
Privacy governance	Complete a UK GDPR/DPA screening and DPIA before launch if the processing is likely high-risk. Maintain lawful-basis mapping, records of processing, data-subject rights workflows and a sub-processor register.
Voice consent & retention	Obtain explicit, separate notice/consent for voice capture where required; define recording/transcript retention, deletion and whether raw audio is stored at all.
Score-claim calibration	Do not present AI-generated Writing/Speaking numeric bands as official or guaranteed. Numeric score estimates require calibration against known official outcomes; otherwise show criterion-level feedback and uncertainty.
AI output reporting	Mobile apps must include an in-app way to flag/report problematic AI-generated content and a moderation/response workflow where store policy requires it.
Clinical boundary	The product teaches communication, language and exam technique. It must not make patient-specific clinical decisions, diagnose, prescribe, or present role-play content as clinical guidance. Genuine clinical questions are redirected to appropriate professional/official clinical sources while the AI may still explain the language or communication task.

Area	Requirement
Learner distress	Respond supportively and non-judgmentally without acting as a counsellor. Never link exam failure to personal worth and never state or imply a pass probability. Maintain a human escalation path owned by Support/Product (named operational owner TO VERIFY); urgent-safety language follows the platform safety runbook.

23. Monetization Architecture: Content Entitlement × AI Entitlement

Commercial rule

Do not hard-wire AI power to a specific course. Content ownership and AI power are two separate axes. A learner can own Nursing Full Course + AI Pro, or Pharmacy content + AI Ultimate, without duplicating product logic.

Axis	Determines	Examples
Content entitlement	What proprietary material the learner may access or ask the AI to use.	Full Course, Crash/Fast Track, Reading-only, Tutor Book, Recalls, profession-specific materials.
AI entitlement	How powerful the assistant is and how much expensive AI usage is included.	Free, Plus, Pro, Ultimate, Enterprise.
AI Credits	Allowance for high-cost assessment/voice/file operations.	Writing assessment, Speaking assessment, deep mock analysis, heavy document/audio analysis.

- Normal navigation should never consume AI Credits.
- Simple grammar/course questions should not consume premium credits unless they trigger an explicitly expensive assessment action.
- Users can purchase extra AI Credits without changing the base AI tier.
- Course bundle discounts may be offered, but the included AI cost must be priced into the bundle.
- Never give perpetual Pro/Ultimate access “for free” with a course unless course pricing covers it.

24. Recommended Launch Pricing and Limits

Pricing objective

Price for a strong perceived-value gap between tiers while protecting net contribution after VAT, store/payment fees and AI variable cost. Initial guardrail: target at least 75% margin on net revenue after channel fees versus AI variable cost; final contribution must also include support, refunds/chargebacks and hosting. Tune limits using measured beta usage.

Tier	Launch price	Base text allowance	What must feel unlocked	Commercial role
FREE	£0	20 messages / 30 days	Mini diagnostic + plan preview; navigation; basic public exam Q&A; one short speaking demo.	Lead generation only.
PLUS	£7.99 / month	~400 text messages	Basic personalised plan; 30-day memory; purchased-content Q&A; limited files/voice; basic Error Bank.	Low-cost conversion tier.
PRO	£14.99 / month	~1,200 text messages	Adaptive plan; long-term memory; Error DNA; advanced course-aware tutor; deeper analytics; voice/speaking; AI beside videos.	Recommended / best value.
ULTIMATE	£26.99 / month	~3,000 text messages / fair use	Continuous AI Mentor; advanced fingerprint; proactive coaching; full multimodal; largest included AI Credit wallet; priority reasoning and premium assessment/voice access through credits.	Premium outcome tier.

25. Detailed AI Tier Feature Matrix

Feature	Free	Plus	Pro	Ultimate
Public/navigation Q&A	✓	✓	✓	✓
Arabic + English + code-switching	✓	✓	✓	✓
Mini onboarding diagnostic	✓	✓	✓	✓
Full personalised profile	Limited	Basic	✓	✓ Advanced
Study plan	Preview	Basic weekly	Adaptive daily	Continuous adaptive
Long-term learning memory	-	30 days	✓	✓ Advanced
Error DNA	-	Basic	✓	✓ Advanced
Learning Fingerprint	-	-	✓	✓ Advanced
Course-aware answers	Preview only	Owned content	Owned content	Owned content
Dr Hesham Rule Book depth	Locked preview	Owned scope	Full owned scope	Full owned scope
What should I do now?	Basic	✓	✓	Next-best-action engine
Generated drills	Very limited	Moderate	High	Very high / fair use
File/image analysis	-	5 / month	25 / month	100 / month
Voice conversation	3-min demo	AI Credits	AI Credits	AI Credits; largest wallet
Speaking simulation	Demo	Basic	Advanced	Full + difficult modes
Writing deep assessment	-	AI Credits	AI Credits	AI Credits + higher included wallet
Speaking deep assessment	-	AI Credits	AI Credits	AI Credits + higher included wallet
Readiness/mastery analytics	Preview	Basic	Advanced	Advanced + predictive trends
Auto replanning	-	Weekly	Daily	Continuous
Proactive coaching	-	-	Limited	✓
AI beside videos/timestamps	-	Limited	✓	✓
Tutor handoff summary	-	-	✓	✓
Priority reasoning/model tier	-	Standard	Higher	Highest appropriate

Feature	Free	Plus	Pro	Ultimate
New experimental features	-	-	Selected	First access
Included monthly AI Credits	0	2 provisional*	8 provisional*	20 provisional*

* Included-credit counts are commercial placeholders only. Production launch wallets must be re-derived from the approved target cost per AI Credit and each tier's total AI-cost ceiling after the live cost beta; if the economics do not fit, reduce the wallet or raise the plan price.

26. AI Credits and Top-Up Revenue

Action	AI Credit charge
Full Writing assessment	2
Full Speaking assessment	2
Full Reading exam analysis	1
Full Listening exam analysis	1
Deep cross-subtest mock analysis	TO VERIFY after live cost test
Voice role play / live voice coaching	TO VERIFY after 2-week voice spike; meter by time/cost
Large PDF / extended audio analysis	TO VERIFY after live cost test

Top-up pack	Recommended price	Positioning
5 AI Credits	£5.99	Occasional extra assessments / voice.
15 AI Credits	£13.99	Recommended best-value top-up.
30 AI Credits	£23.99	Heavy exam-prep period / intensive practice.

- AI Credits are separate from normal chat messages.
- Show remaining balance clearly before a chargeable action.
- Ask for confirmation when an action will consume AI Credits.
- Do not charge for failed/technical-error attempts.
- Allow admin-granted promotional actions, with clear expiry and source.

AI Credit migration and accounting rules

- Use one user-facing currency only: AI Credits. Existing unspent credits migrate 1:1; there is no haircut and no second wallet called “Power Actions”.
- Backend should track credit provenance separately (course-granted, subscription-included, purchased top-up, promotional/admin grant) even if the learner sees one total balance.

- Existing assessment charges remain compatible: Writing 2 credits, Speaking 2 credits, Reading 1 credit and Listening 1 credit. Voice conversion is deliberately TO VERIFY until the live voice-cost spike is complete.
- Grandfather existing learners: unspent credits remain valid under their existing entitlement/expiry rules unless a separate migration notice is approved.
- Never deduct for failed technical attempts. Show the exact charge before any credit-consuming action and keep an audit trail of every credit movement.

Cost per AI Credit - mandatory v3 economic control

Target fully loaded variable delivery cost per AI Credit: TO VERIFY in the live cost beta. It must include the model/voice/transcription/file-processing cost attributable to the action, not just the visible model token line item.

- Wallet rule: first reserve a TO VERIFY portion of each tier's AI variable-cost ceiling for ordinary text chat, retrieval, memory, files and other non-credit usage. Only the remaining credit budget may fund included AI Credits. Therefore: included credits × approved target unit cost must never exceed the remaining tier credit budget.
- Action-pricing rule: the expected/p95 cost of a chargeable action must fit inside (credits charged × approved target cost per credit). Voice-heavy or long-media actions may charge more credits than text-only actions with the same user-facing purpose.
- Top-up floor: no credit pack may be sold on a channel if net revenue per credit, after destination tax and channel/payment fees, is below the approved multiple of target cost per credit. Required multiple: TO VERIFY - Finance/Product.
- Credit refund/reversal policy: the ledger must support purchase-level reversal, unused-credit revocation and technical-failure restoration. On cancellation, cooling-off refund or chargeback, treatment of partially consumed purchased credits follows the approved consumer-law/channel policy; exact legal rule TO VERIFY. Subscription-included/course-granted/promotional credits retain their documented provenance and expiry.

No production tier or top-up pack is financially approved until the target cost per AI Credit, redemption mix and channel-specific floor have been validated with live telemetry.

27. Profit Protection and Internal Cost Ceilings

Tier	Selling price	Initial AI variable-cost ceiling / active user / month*	Net-revenue margin guardrail
Plus	£7.99	≤ £1.00	Target ≥75% after VAT/store/payment fees, before support/hosting
Pro	£14.99	≤ £2.20	Target ≥75% after VAT/store/payment fees, before support/hosting
Ultimate	£26.99	≤ £4.20	Target ≥75% after VAT/store/payment fees, before support/hosting

Important

These are initial internal control ceilings, not vendor promises. The operational rule is stricter: AI variable cost must also stay within the approved percentage of NET revenue after tax and channel fees. Voice is metered through AI Credits so a single heavy user cannot consume the full plan margin.

- Route simple navigation/classification to low-cost models or deterministic platform data.
- Use stronger models only when deep reasoning, assessment or complex planning requires them.
- Cache/reuse stable approved answers and resource maps when safe.
- Summarise long conversation history into structured learner memory instead of repeatedly sending entire raw histories.
- Set tier-specific token, file-size, audio-duration and rate limits.
- Track cost per feature, user, tier, profession and cohort daily.
- Automatic margin alert if average tier cost crosses its ceiling.
- Temporary throttling or fair-use controls for abusive/automation-like usage.

Emergency cost and product kill switches

Named authority: Founder/Product Owner may trigger the emergency control; AI Engineering executes and logs it. Delegated backup owner: TO VERIFY.

- Disable or cap live voice immediately without disabling text tutoring.
- Freeze new AI Credit consumption while preserving balances during a billing/ledger incident.
- Temporarily suspend a tier, action type, vendor route or file/audio workload if unit economics or abuse cross the approved threshold.
- Roll back the latest knowledge release or prompt/model configuration if quality, leakage or safety regression is detected.
- Every kill-switch event requires an incident record, customer-impact note, recovery criteria and explicit re-enable approval.

28. Illustrative Revenue Economics

The legacy table below is retained only as an illustrative gross-billings snapshot. It is NOT an LTV, CAC, margin or contribution model. The authoritative v3.0 economics are in Sections 38-39, which model VAT, channel/payment fees, AI variable cost, support/refund provisions, customer lifetime, churn and reactivation.

Active paid users	Illustrative GROSS subscription billings / month*	Not LTV / not net contribution
100	~£1,494	Before VAT, stores, payment fees, AI, support
250	~£3,735	Before VAT, stores, payment fees, AI, support
500	~£7,470	Before VAT, stores, payment fees, AI, support

Active paid users	Illustrative GROSS subscription billings / month*	Not LTV / not net contribution
1,000	~£14,940	Before VAT, stores, payment fees, AI, support

*Gross-billings illustration only. Do not use this table for margin, LTV or CAC decisions. See Sections 38-39 for the net-revenue waterfall and cohort model.

29. Course Bundles and AI Add-Ons

Recommended rule

Existing course/package ownership unlocks the proprietary content the AI may discuss, but does not automatically unlock the strongest AI tier. AI remains an upsell and recurring-revenue product.

Buyer type	Default AI access	Recommended upsell
Free registered user	Free	Plus / Pro / Ultimate
Tutor Book / Recalls buyer	Free + eligible content preview/navigation	Plus or Pro
Single-subtest / Reading-only buyer	Free or time-limited basic companion if priced into purchase	Plus / Pro
Crash / Fast Track buyer	Course-aware limited AI only if bundle price covers usage	Pro / Ultimate
Full Course buyer	Course-aware limited AI or short paid-in bundle period	Pro / Ultimate
Premium “Ultimate Course” bundle	Ultimate for a defined 60/90-day exam-prep window	Renew Ultimate monthly or top up

Optional course-checkout AI add-on	1 month	3 months
Plus add-on	£6.99	£18.99
Pro add-on	£12.99	£34.99
Ultimate add-on	£23.99	£64.99

Bundle discounts are intentionally lower than standalone monthly prices, but the AI period must have a defined expiry. Do not provide perpetual premium AI access by accident.

30. Upgrade UX, Paywalls and Conversion

- Anonymous visitors receive approximately 3 demo interactions, then are asked to create a free account to continue.
- Registered Free users receive approximately 20 messages per rolling 30-day period and see a visible usage counter.
- Locked premium features should show a personalised preview, not a dead error message.
- Upgrade cards must show the exact tier, monthly price, 3-5 benefits relevant to the action, and one clear CTA.
- Contextual paywall: a user asking for a 30-day adaptive plan sees the plan upgrade card; a user attempting voice role play sees the Speaking/Pro card.
- Checkout should preserve the current chat. After successful payment, the feature unlocks immediately and the same conversation resumes.
- Show “why upgrade” using the candidate’s actual need: exam date, weak skill, requested action, or usage limit.
- Never repeatedly interrupt paid users with irrelevant upsells.
- Allow upgrade, downgrade, cancel, renew and top-up management without support tickets where possible.
- Use local-currency display/regional promotions where commercially appropriate, while maintaining a clear GBP base price.

31. Growth, Lead Generation and SEO Intelligence

- Use the public AI demo as a lead-capture engine: profession, exam date and core problem become high-value segmentation data after consent/registration.
- Mine anonymised query trends to discover high-intent public topics and content gaps.
- Create an admin SEO/content queue from frequently asked questions; human/editorial approval is required before public publishing.
- Generate internal-link suggestions between public guides, course pages and relevant resources.
- Create FAQ/topic clusters from real learner demand rather than mass-producing low-value AI pages.
- At positive milestones/result success, optionally request a review/testimonial or offer a referral link - never during moments of frustration.
- Track free→paid conversion by paywall trigger, exam type, profession, traffic source and first “wow” moment.

Owned-audience launch sequence - v3 growth correction

The business states an existing owned reach of 52,000+ healthcare professionals across Telegram, Facebook, WhatsApp and YouTube. Treat this as the primary Stage 1 beta/launch pool, not as paid acquisition. Current channel counts and deduplicated reach: TO VERIFY from analytics before launch.

Sequence	Audience / channel	Launch purpose	Conversion target / CAC treatment
1	Existing active/paying learners and recent course buyers	Highest-intent beta; validate grounding, navigation, pricing and credit behavior	Conversion TO VERIFY; media CAC near zero but staff/platform cost must still be recorded.
2	Exam-soon warm community segments on WhatsApp / Telegram	Offer targeted AI study-plan and tutor demo based on exam date/profession	Conversion TO VERIFY by channel.
3	YouTube / Facebook owned audience	Broader education + demo acquisition with clear free-to-paid funnel	Conversion TO VERIFY; track source-to-paid cohort.
4	Public SEO / referral traffic	Cold organic acquisition after warm funnel is stable	Separate cohort economics.
5	Paid cold acquisition	Scale only after the CAC ceiling and payback rule are proven	Do not blend with warm/owned CAC.

32. Multi-Exam Expansion: IELTS, PTE, TOEFL and Beyond

Architecture requirement

Do not encode OET as the only possible exam. The product must treat “Exam” as a modular knowledge and assessment pack with versions, modules, task types, scoring logic, content sources and entitlement rules.

Dimension	Required model
Exam	OET, IELTS, PTE, TOEFL, future exams/curricula.
Version	Effective date, retired/current format, delivery mode.
Module/Subtest	Exam-specific skills and sections.
Task type	Question/task-specific rules and assessment logic.
Profession/track	OET profession, IELTS Academic/General, institution-specific tracks, etc.
Rule Book	Approved teaching methodology for that exam/task.
Content pack	Videos, lessons, exercises, mocks, vocab and support.
Candidate mastery	Per-exam mastery that can reuse transferable skills when sensible.

- Cross-exam skill transfer: preserve vocabulary/listening/grammar strengths when a learner moves exams, but rebuild task-specific strategy.
- Goal-based exam advisor can compare accepted exams for a user's country/regulator only using current verified sources.
- Separate current official requirements from internal recommendation and never present stale regulator rules as fact.

33. B2B, White-Label and Enterprise Product

- Multi-tenant institution model with strict data/knowledge separation.
- White-label logo, colour, assistant name/persona and institution-specific welcome behavior.
- Institution uploads its curriculum, policies, FAQs and approved teaching methodology.
- Custom learner/tutor/admin roles and permissions.
- Seat management, active-user limits, usage budgets and per-tenant AI cost reporting.
- Institution analytics, cohort mastery, engagement, at-risk learners and content-gap reporting.
- Tutor copilot, candidate briefs and institution-specific escalation rules.
- Custom AI tier/feature restrictions by institution contract.
- SSO, API, webhooks and LMS integrations as enterprise capabilities.
- Audit logs, custom retention policies and enterprise support.
- Optional institution-specific public or internal assistant while using the same core AI platform.

Indicative future B2B commercial model	Recommendation
Small institution	Platform minimum + per-active-learner fee.
Medium/large institution	Annual contract with seat/usage band and support level.
White label	Setup/branding fee + recurring platform/usage fee.
API/enterprise integration	Custom contract based on usage, SLA and integration scope.

34. Business and AI Operations Dashboard

- Monthly recurring revenue by AI tier
- Free→paid conversion rate
- Upgrade trigger conversion
- Paid churn and downgrade rate
- ARPU and top-up revenue
- AI cost per active user/tier
- Gross margin by tier
- Cost by feature (voice, writing, speaking, files, text)
- Messages and AI Credits/user
- Retention at 7/30/90 days
- Plan adherence and learning engagement
- Support deflection rate
- Tutor escalation rate
- Thumbs up/down and answer quality
- Low-confidence/unknown-answer rate
- Content gaps and most-asked topics
- Profession/exam/tier usage mix
- B2B tenant cost/revenue/seat utilisation

Commercial alerting

Admin should be warned automatically if a tier's average AI cost breaches the internal ceiling, if abusive usage emerges, or if a specific feature becomes structurally unprofitable.

35. Complete Feature Inventory / Traceability Checklist

The following inventory is intended as a “nothing silently disappears” checklist for product, design and development. Items may launch in phases, but each should be explicitly accepted, deferred or rejected - never omitted accidentally.

Identity & onboarding

ID	Requirement	Decision
F-001	Anonymous demo mode	☐ Accept ☐ Defer ☐ Reject
F-002	Registered Free mode	☐ Accept ☐ Defer ☐ Reject
F-003	Profession selection	☐ Accept ☐ Defer ☐ Reject
F-004	Exam/version selection	☐ Accept ☐ Defer ☐ Reject
F-005	Exam date and target	☐ Accept ☐ Defer ☐ Reject

ID	Requirement	Decision
F-006	Country/regulator goal	☐ Accept ☐ Defer ☐ Reject
F-007	Previous result capture	☐ Accept ☐ Defer ☐ Reject
F-008	Screenshot score import	☐ Accept ☐ Defer ☐ Reject
F-009	Study availability	☐ Accept ☐ Defer ☐ Reject
F-010	Language/code-switch preference	☐ Accept ☐ Defer ☐ Reject
F-011	Teaching-style preference	☐ Accept ☐ Defer ☐ Reject
F-012	Multiple historical exam journeys	☐ Accept ☐ Defer ☐ Reject

Knowledge & grounding

ID	Requirement	Decision
F-013	All approved Rule Books	☐ Accept ☐ Defer ☐ Reject
F-014	All relevant session transcripts	☐ Accept ☐ Defer ☐ Reject
F-015	Writing workshops	☐ Accept ☐ Defer ☐ Reject
F-016	Speaking workshops	☐ Accept ☐ Defer ☐ Reject
F-017	Correction sessions	☐ Accept ☐ Defer ☐ Reject
F-018	Reading/Listening materials	☐ Accept ☐ Defer ☐ Reject
F-019	Tutor Book	☐ Accept ☐ Defer ☐ Reject
F-020	Dictionaries/common-word lists	☐ Accept ☐ Defer ☐ Reject
F-021	Recalls/practice banks	☐ Accept ☐ Defer ☐ Reject
F-022	Video timestamp index	☐ Accept ☐ Defer ☐ Reject
F-023	Platform navigation map	☐ Accept ☐ Defer ☐ Reject
F-024	Support/FAQ knowledge	☐ Accept ☐ Defer ☐ Reject
F-025	Official current exam sources	☐ Accept ☐ Defer ☐ Reject
F-026	Source hierarchy	☐ Accept ☐ Defer ☐ Reject
F-027	Content versioning	☐ Accept ☐ Defer ☐ Reject
F-028	Teacher override	☐ Accept ☐ Defer ☐ Reject
F-029	Conflict detection	☐ Accept ☐ Defer ☐ Reject

Planning & memory

ID	Requirement	Decision
F-030	Initial diagnostic	☐ Accept ☐ Defer ☐ Reject
F-031	Daily plan	☐ Accept ☐ Defer ☐ Reject
F-032	Weekly plan	☐ Accept ☐ Defer ☐ Reject
F-033	30/60/90-day plans	☐ Accept ☐ Defer ☐ Reject
F-034	14/7/3-day plans	☐ Accept ☐ Defer ☐ Reject
F-035	Exam-eve plan	☐ Accept ☐ Defer ☐ Reject
F-036	Missed-day replanning	☐ Accept ☐ Defer ☐ Reject
F-037	Shift-worker planning	☐ Accept ☐ Defer ☐ Reject
F-038	Travel-aware replanning	☐ Accept ☐ Defer ☐ Reject
F-039	What should I do now?	☐ Accept ☐ Defer ☐ Reject
F-040	Next-best-action engine	☐ Accept ☐ Defer ☐ Reject
F-041	Conversation memory	☐ Accept ☐ Defer ☐ Reject
F-042	Learning memory	☐ Accept ☐ Defer ☐ Reject
F-043	Journey memory	☐ Accept ☐ Defer ☐ Reject
F-044	Error DNA	☐ Accept ☐ Defer ☐ Reject
F-045	Learning Fingerprint	☐ Accept ☐ Defer ☐ Reject
F-046	Spaced reinforcement	☐ Accept ☐ Defer ☐ Reject
F-047	Memory controls/export/delete	☐ Accept ☐ Defer ☐ Reject

Tutoring

ID	Requirement	Decision
F-048	Quick answer	☐ Accept ☐ Defer ☐ Reject
F-049	Detailed tutor	☐ Accept ☐ Defer ☐ Reject
F-050	Socratic tutor	☐ Accept ☐ Defer ☐ Reject
F-051	Examiner mode	☐ Accept ☐ Defer ☐ Reject
F-052	Coach mode	☐ Accept ☐ Defer ☐ Reject
F-053	Dr Hesham mode	☐ Accept ☐ Defer ☐ Reject
F-054	Arabic explanation	☐ Accept ☐ Defer ☐ Reject
F-055	English-only mode	☐ Accept ☐ Defer ☐ Reject

ID	Requirement	Decision
F-056	Adaptive drill generator	☐ Accept ☐ Defer ☐ Reject
F-057	Micro-lessons	☐ Accept ☐ Defer ☐ Reject
F-058	Grammar tutor	☐ Accept ☐ Defer ☐ Reject
F-059	Vocabulary brain	☐ Accept ☐ Defer ☐ Reject
F-060	Writing guided mode	☐ Accept ☐ Defer ☐ Reject
F-061	Writing hint mode	☐ Accept ☐ Defer ☐ Reject
F-062	Writing correction	☐ Accept ☐ Defer ☐ Reject
F-063	Writing compare/rewrite	☐ Accept ☐ Defer ☐ Reject
F-064	Writing scoring/criteria feedback	☐ Accept ☐ Defer ☐ Reject
F-065	Speaking voice role play	☐ Accept ☐ Defer ☐ Reject
F-066	Speaking difficult-patient modes	☐ Accept ☐ Defer ☐ Reject
F-067	Speaking pronunciation/fluency analysis	☐ Accept ☐ Defer ☐ Reject
F-068	Reading Part A/B/C coach	☐ Accept ☐ Defer ☐ Reject
F-069	Listening Part A/B/C coach	☐ Accept ☐ Defer ☐ Reject
F-070	Confidence-vs-accuracy analysis	☐ Accept ☐ Defer ☐ Reject
F-071	Why did my score change?	☐ Accept ☐ Defer ☐ Reject

Exam & analytics

ID	Requirement	Decision
F-072	Full mock mode	☐ Accept ☐ Defer ☐ Reject
F-073	Practice vs Exam mode separation	☐ Accept ☐ Defer ☐ Reject
F-074	Computer-based rehearsal	☐ Accept ☐ Defer ☐ Reject
F-075	Test-day rehearsal	☐ Accept ☐ Defer ☐ Reject
F-076	Final 24-hours mode	☐ Accept ☐ Defer ☐ Reject
F-077	Result-day assistant	☐ Accept ☐ Defer ☐ Reject
F-078	Resit recovery plan	☐ Accept ☐ Defer ☐ Reject
F-079	Readiness score	☐ Accept ☐ Defer ☐ Reject
F-080	Mastery map	☐ Accept ☐ Defer ☐ Reject
F-081	Trend analytics	☐ Accept ☐ Defer ☐ Reject

ID	Requirement	Decision
F-082	Timing analytics	☐ Accept ☐ Defer ☐ Reject
F-083	Personal bests	☐ Accept ☐ Defer ☐ Reject
F-084	Anonymous cohort benchmarking	☐ Accept ☐ Defer ☐ Reject
F-085	Weekly progress report	☐ Accept ☐ Defer ☐ Reject

Multimodal & context

ID	Requirement	Decision
F-086	Image/screenshot understanding	☐ Accept ☐ Defer ☐ Reject
F-087	PDF understanding	☐ Accept ☐ Defer ☐ Reject
F-088	Audio analysis	☐ Accept ☐ Defer ☐ Reject
F-089	Handwritten note understanding	☐ Accept ☐ Defer ☐ Reject
F-090	Score report extraction	☐ Accept ☐ Defer ☐ Reject
F-091	Current-page awareness	☐ Accept ☐ Defer ☐ Reject
F-092	Current-question awareness	☐ Accept ☐ Defer ☐ Reject
F-093	Current-video awareness	☐ Accept ☐ Defer ☐ Reject
F-094	Timestamp-aware help	☐ Accept ☐ Defer ☐ Reject
F-095	Writing selection awareness	☐ Accept ☐ Defer ☐ Reject
F-096	Speaking-session context	☐ Accept ☐ Defer ☐ Reject
F-097	Cross-device conversation continuity	☐ Accept ☐ Defer ☐ Reject

Actions & navigation

ID	Requirement	Decision
F-098	Deep-link resource open	☐ Accept ☐ Defer ☐ Reject
F-099	Open exact timestamp	☐ Accept ☐ Defer ☐ Reject
F-100	Start recommended practice	☐ Accept ☐ Defer ☐ Reject
F-101	Continue last activity	☐ Accept ☐ Defer ☐ Reject
F-102	Save note	☐ Accept ☐ Defer ☐ Reject
F-103	Save vocabulary	☐ Accept ☐ Defer ☐ Reject
F-104	Add to plan	☐ Accept ☐ Defer ☐ Reject
F-105	Set reminder	☐ Accept ☐ Defer ☐ Reject

ID	Requirement	Decision
F-106	Open workshop	☐ Accept ☐ Defer ☐ Reject
F-107	Create support request	☐ Accept ☐ Defer ☐ Reject
F-108	Tutor handoff summary	☐ Accept ☐ Defer ☐ Reject
F-109	Report wrong answer	☐ Accept ☐ Defer ☐ Reject
F-110	Show allowances/credits	☐ Accept ☐ Defer ☐ Reject
F-111	Contextual upgrade checkout	☐ Accept ☐ Defer ☐ Reject
F-112	Resume chat after purchase	☐ Accept ☐ Defer ☐ Reject

Proactive & engagement

ID	Requirement	Decision
F-113	Exam countdown	☐ Accept ☐ Defer ☐ Reject
F-114	Inactivity risk nudges	☐ Accept ☐ Defer ☐ Reject
F-115	Weak-subtest reminder	☐ Accept ☐ Defer ☐ Reject
F-116	New relevant content alert	☐ Accept ☐ Defer ☐ Reject
F-117	Calendar integration	☐ Accept ☐ Defer ☐ Reject
F-118	Push/email/app progress summaries	☐ Accept ☐ Defer ☐ Reject
F-119	Quiet hours	☐ Accept ☐ Defer ☐ Reject
F-120	Streaks/milestones	☐ Accept ☐ Defer ☐ Reject
F-121	Adaptive gamification	☐ Accept ☐ Defer ☐ Reject
F-122	Referral/review prompt at positive moments	☐ Accept ☐ Defer ☐ Reject

Admin/tutor

ID	Requirement	Decision
F-123	Tutor pre-session brief	☐ Accept ☐ Defer ☐ Reject
F-124	Tutor post-session notes	☐ Accept ☐ Defer ☐ Reject
F-125	Admin aggregate AI queries	☐ Accept ☐ Defer ☐ Reject
F-126	Quality dashboard	☐ Accept ☐ Defer ☐ Reject
F-127	Hallucination/low-confidence queue	☐ Accept ☐ Defer ☐ Reject
F-128	Content-gap dashboard	☐ Accept ☐ Defer ☐ Reject

ID	Requirement	Decision
F-129	Teaching-gap dashboard	☐ Accept ☐ Defer ☐ Reject
F-130	Rule approval/version/rollback	☐ Accept ☐ Defer ☐ Reject
F-131	Content Studio extraction proposals	☐ Accept ☐ Defer ☐ Reject
F-132	Revenue/cost dashboard	☐ Accept ☐ Defer ☐ Reject
F-133	Entitlement management	☐ Accept ☐ Defer ☐ Reject
F-134	Promotional AI Credits	☐ Accept ☐ Defer ☐ Reject

Commercial

ID	Requirement	Decision
F-135	Free message cap	☐ Accept ☐ Defer ☐ Reject
F-136	Plus tier	☐ Accept ☐ Defer ☐ Reject
F-137	Pro tier	☐ Accept ☐ Defer ☐ Reject
F-138	Ultimate tier	☐ Accept ☐ Defer ☐ Reject
F-139	AI Credits	☐ Accept ☐ Defer ☐ Reject
F-140	Top-up packs	☐ Accept ☐ Defer ☐ Reject
F-141	Usage counter	☐ Accept ☐ Defer ☐ Reject
F-142	Contextual paywall	☐ Accept ☐ Defer ☐ Reject
F-143	Course × AI entitlement matrix	☐ Accept ☐ Defer ☐ Reject
F-144	Standalone AI subscription	☐ Accept ☐ Defer ☐ Reject
F-145	Course AI add-on	☐ Accept ☐ Defer ☐ Reject
F-146	Monthly billing	☐ Accept ☐ Defer ☐ Reject
F-147	Optional annual billing	☐ Accept ☐ Defer ☐ Reject
F-148	Upgrade/downgrade/cancel	☐ Accept ☐ Defer ☐ Reject
F-149	Fair-use/rate limits	☐ Accept ☐ Defer ☐ Reject
F-150	Regional currency display	☐ Accept ☐ Defer ☐ Reject
F-151	Cost ceiling alerts	☐ Accept ☐ Defer ☐ Reject

Trust & platform

ID	Requirement	Decision
F-152	Official-vs-methodology separation	☐ Accept ☐ Defer ☐ Reject

ID	Requirement	Decision
F-153	Anti-hallucination behavior	☐ Accept ☐ Defer ☐ Reject
F-154	Entitlement-safe retrieval	☐ Accept ☐ Defer ☐ Reject
F-155	Academic integrity controls	☐ Accept ☐ Defer ☐ Reject
F-156	Sensitive upload warning	☐ Accept ☐ Defer ☐ Reject
F-157	Privacy controls	☐ Accept ☐ Defer ☐ Reject
F-158	Data export/delete	☐ Accept ☐ Defer ☐ Reject
F-159	Audit logs	☐ Accept ☐ Defer ☐ Reject
F-160	Accessibility - WCAG 2.2 AA across learner-facing web, web-app and native-app surfaces	☐ Accept ☐ Defer ☐ Reject
F-161	Arabic RTL support	☐ Accept ☐ Defer ☐ Reject
F-162	iOS	☐ Accept ☐ Defer ☐ Reject
F-163	Android	☐ Accept ☐ Defer ☐ Reject
F-164	Windows	☐ Accept ☐ Defer ☐ Reject
F-165	macOS	☐ Accept ☐ Defer ☐ Reject
F-166	Website	☐ Accept ☐ Defer ☐ Reject
F-167	Web app	☐ Accept ☐ Defer ☐ Reject

Expansion & B2B

ID	Requirement	Decision
F-168	IELTS pack	☐ Accept ☐ Defer ☐ Reject
F-169	PTE pack	☐ Accept ☐ Defer ☐ Reject
F-170	TOEFL pack	☐ Accept ☐ Defer ☐ Reject
F-171	Future exam packs	☐ Accept ☐ Defer ☐ Reject
F-172	Exam-version engine	☐ Accept ☐ Defer ☐ Reject
F-173	Cross-exam skill transfer	☐ Accept ☐ Defer ☐ Reject
F-174	B2B multi-tenancy	☐ Accept ☐ Defer ☐ Reject
F-175	White-label persona	☐ Accept ☐ Defer ☐ Reject
F-176	Institution knowledge upload	☐ Accept ☐ Defer ☐ Reject

ID	Requirement	Decision
F-177	Institution roles	☐ Accept ☐ Defer ☐ Reject
F-178	Seat/usage budgets	☐ Accept ☐ Defer ☐ Reject
F-179	Institution analytics	☐ Accept ☐ Defer ☐ Reject
F-180	SSO	☐ Accept ☐ Defer ☐ Reject
F-181	API/webhooks	☐ Accept ☐ Defer ☐ Reject
F-182	LMS integration	☐ Accept ☐ Defer ☐ Reject
F-183	Custom retention	☐ Accept ☐ Defer ☐ Reject
F-184	Enterprise audit/support	☐ Accept ☐ Defer ☐ Reject

36. Recommended Release Sequence

Stage	Business objective	Minimum capability set
Stage 1 - Monetisable OET Core	Start charging quickly with a safe, useful OET companion.	Approx. 15 core capabilities only: onboarding; one-profession entitlement-safe retrieval; platform navigator/deep links; basic study plan; “What should I do now?”; basic memory; existing Writing assessment re-plumbed; Free/Plus/Pro; usage counter; contextual paywall/checkout; source-authority separation; privacy controls; exfiltration defence; cost dashboard; support/feedback loop.
Stage 2 - Learning Moat	Create value a generic chat tool cannot reproduce.	Full approved Rule Book/material grounding; Error DNA; adaptive plan; Reading/Listening/Writing tutors; video/timestamp intelligence; deeper analytics; unified AI Credits; content-ops pipeline.
Stage 3 - Voice & Ultimate Mentor	Only launch Ultimate when its premium promise is real.	Arabic/English voice after feasibility gate; Speaking simulator; Learning Fingerprint; proactive coaching; continuous replanning; multimodal; tutor collaboration.
Stage 4 - Multi-Exam	Expand TAM without rebuilding core.	IELTS/PTE/TOEFL packs; version engine; task-specific rules/evaluation; cross-exam transfer.
Stage 5 - B2B Platform	Create institution/enterprise revenue.	Multi-tenancy; white label; institution analytics; roles; SSO/API/LMS; budgets/contracts; regional data-residency options.

Launch principle

Phasing is allowed; silent omission is not. Every item in the traceability checklist should have an explicit product decision and owner before final scope sign-off.

37. Final Acceptance Criteria

- A learner can register, identify profession/exam/date/target and receive a meaningful personalised recommendation quickly.
- The AI can answer from all approved eligible OET Rule Books and materials without leaking locked content.
- The AI differentiates official exam facts from Dr Hesham teaching methodology.
- The AI can explain where a resource is and open it directly when a deep link exists.
- The same learner identity, memory and plan follow the user across supported platforms.
- The study plan adapts after scores, missed days and exam-date changes.
- Free usage has a hard commercial cap and every locked premium feature has a clear contextual upgrade path.
- Plus, Pro and Ultimate have visibly different outcomes, not merely different message counts.
- Ultimate behaves like a proactive mentor with continuous preparation management.
- Expensive features are separately metered and top-ups are available.
- Admin can see revenue, AI cost, margin, quality and content gaps.
- Admins can correct/approve teaching knowledge without uncontrolled AI self-modification.
- Current exam/version information can be updated without rebuilding the assistant.
- The architecture can add IELTS, PTE, TOEFL and B2B knowledge packs later.
- Privacy, access, source authority, academic integrity and tenant isolation are enforced as product requirements.

Free demonstrates the magic. Pro teaches deeply. Ultimate manages the entire learning journey.

Working user-facing identity

Talk to Jana / Talk to Sami

Commercial north star

Every premium interaction must create enough learner value to justify payment, while every high-cost AI action remains measurable, capped and profitable.

End of Core Product Vision - v3.0 operational, economics, architecture and sign-off sections follow

38. Consolidated Unit Economics, Channel Fees and Regional Pricing

Commercial rule

Profitability is measured from NET revenue, not sticker price. Every channel must pass: Gross price - destination tax - store/payment fees = Net revenue - AI variable cost - support/human cost - refund/chargeback provision = contribution.

Verified channel assumptions used for the UK examples below (30 Aug 2026)

- UK standard VAT rate: 20% for most goods/services; exact digital-service tax treatment and customer location rules must be confirmed by the business accountant.
- Apple: the Small Business Program advertises a 15% reduced commission on paid apps and in-app purchases for eligible developers. Exact program/enrolment/storefront terms are TO VERIFY before launch.
- Google Play: for UK/EEA/US transactions from 30 June 2026, auto-renewing subscriptions within the first US\$1M annual earnings are listed at 10% service fee plus 5% billing fee when Google Play Billing applies. Eligibility/status is TO VERIFY.
- Stripe UK benchmark: standard UK cards 1.5% + £0.20; international cards 3.15% + £0.20, with possible FX fees. If a different processor is actually used, replace the benchmark with its live contract.

Tier	Gross UK price	Ex-VAT	Web net*	iOS/Google illustrative net**	AI ceiling	Contribution after AI (web / store)
Plus	£7.99	£6.66	£6.34	£5.66	£1.00	£5.34 / £4.66
Pro	£14.99	£12.49	£12.07	£10.62	£2.20	£9.87 / £8.42
Ultimate	£26.99	£22.49	£21.89	£19.12	£4.20	£17.69 / £14.92

*Web example uses Stripe standard UK-card benchmark. **Illustrative 15% store deduction after VAT for an eligible small-business subscription case; exact store terms vary and must be checked. These figures still exclude hosting, support, refunds/chargebacks and corporate tax.

Billing options

Tier	Monthly	3-month prepaid	Annual
Plus	£7.99	£21.99	£79.99
Pro	£14.99	£39.99	£149.99
Ultimate	£26.99	£72.99	£269.99

For OET, the 3-month prepaid option is likely more natural than forcing annual billing because exam preparation often self-terminates after the test. Annual remains useful for long-horizon learners, cross-exam users and future B2B. This is a commercial design decision, not an external market fact.

Regional purchasing-power framework

Band	Indicative markets	Price rule	Payment / fraud control
A - Base	UK, Europe, Gulf and other high-ARPU markets	100% GBP base price	Local currency display; channel tax/store rules apply.
B - Regional	South Asia, Philippines, Nigeria and similar markets	Discount level TO VERIFY by willingness-to-pay test	Require billing-country/store-region consistency; do not use IP alone.
C - Egypt / selected PPP markets	Egypt and other approved low-PPP markets	Discount level TO VERIFY; protect a minimum contribution floor	Prefer low-cost local rails where available; do not offer a store price that breaks the contribution floor.

- Candidate local rails to evaluate include Fawry, InstaPay and Vodafone Cash in Egypt, plus processor-supported local methods in other markets. Availability, merchant settlement, fees and chargeback rules are TO VERIFY before being advertised.
- Web checkout and in-app purchase may have different prices if permitted by applicable store rules. Steering language must be region/storefront compliant; never assume the US external-link rule applies globally.
- Use a hard minimum contribution floor. If a local price or store fee would push the plan below the floor, reduce included expensive usage, use AI Credits, raise price, or do not sell that tier through that channel.

AI Credit top-up channel waterfall

Credit packs are consumable digital purchases. Apply the same destination-tax and channel-fee logic as subscriptions, then test net revenue per credit against the approved cost floor.

30-credit pack example	Gross	After 20% UK VAT	After channel/payment fee	Net revenue / credit
Web - illustrative UK card*	£23.99	£19.99	~£19.43	~£0.65
iOS / Google - illustrative 15% store	£23.99	£19.99	£16.99	£0.57

* Web figure uses the v2 Appendix C illustrative Stripe UK-card fee basis; actual processor/card/FX fees are TO VERIFY per checkout channel.

- Floor rule: if channel net revenue per credit falls below the approved multiple of target delivery cost per credit, raise the price, reduce the pack size/benefit, change the channel, or withhold that pack from that channel. Never subsidise a structurally loss-making credit pack to preserve visual price parity.

39. Cohort Economics, Churn, LTV and CAC

Exam-prep is a short-lifetime product. Revenue planning must therefore use cohorts, not only a monthly subscriber snapshot. The model below uses the v2 UK-web benchmark and the AI ceilings as a conservative planning example; actual cohort values must be replaced by live data.

Tier	Monthly contribution benchmark*	2-month LTV	3-month LTV	4-month LTV
Plus	£5.34	£10.68	£16.02	£21.35
Pro	£9.87	£19.73	£29.60	£39.47
Ultimate	£17.69	£35.37	£53.06	£70.75
Blended 35/45/20 mix	£9.85	£19.69	£29.54	£39.38

*Before support/human cost, hosting, refunds/chargebacks and corporate tax. Mobile-store cohorts will have lower net revenue and must be modelled separately.

- Model 2-, 3- and 4-month average customer lifetime scenarios separately. Do not assume indefinite subscriptions; passing the exam is a natural cancellation event.
- Set a target LTV:CAC ratio and paid-acquisition payback period only after the first reliable paid cohort exists. Current target values: TO VERIFY - owner: Growth/Finance.
- Track voluntary churn separately from involuntary churn (failed payment, expired card, processor decline). Implement dunning/retry flows and measure recovery rate.
- Create a refund and chargeback reserve line in the P&L model. Rate: TO VERIFY from actual processor history.
- Model resit candidates as the primary reactivation cohort: a learner who fails or needs one subtest again should be offered a resit plan and reactivation path, not treated as a completely new acquisition.

Warm vs cold acquisition economics

Model the stated 52,000+ owned audience separately from cold acquisition. Owned/warm cohorts should carry channel/staff/CRM cost but not be burdened with paid-media CAC that was not incurred.

- Interim cold paid-acquisition guardrail: use the current blended 3-month contribution LTV of £29.54 with a working 3:1 LTV:CAC rule, giving a maximum paid CAC of approximately £9.85 and roughly one month of blended contribution payback. This is a temporary policy guardrail, not a measured result; replace it after the first reliable paid cohort.
- Track warm conversion, cold-organic conversion and paid-cold conversion as separate cohorts. Do not average them into one conversion rate until channel-level economics are visible.

40. Technical Architecture Requirements

Layer	Requirement	Acceptance evidence
Retrieval / RAG	Chunk by pedagogical meaning and document structure, not arbitrary fixed text only. Metadata must include exam, version, profession, subtest, content entitlement, source authority, source ID and timestamp/page. Use hybrid lexical + vector retrieval with reranking where it improves accuracy.	Labelled retrieval test set demonstrates acceptable recall/precision; targets TO VERIFY.
Knowledge authority	Source hierarchy must prefer approved Dr Hesham Rule Books/materials for teaching method and verified official sources for official rules. Conflicts are surfaced, not silently blended.	Golden questions show correct authority selection.
Model routing	Route navigation, classification and deterministic lookups to cheap/fast paths; reserve stronger reasoning for planning, difficult tutoring and assessment.	Per-intent cost and latency dashboard.
Latency	Define separate budgets for page Q&A, plan generation, file analysis and live voice. Voice turn-taking requires a dedicated low-latency target; exact threshold TO VERIFY in the feasibility spike.	P50/P95 latency monitoring by surface.
Caching	Cache stable platform maps, approved source snippets and repeated public facts safely. Use prompt/prefix caching where vendor-supported. Never cache across entitlement/tenant boundaries.	Cache hit rate + no cross-user leakage.
Memory compaction	Summarise long chat into structured learner memory and event history instead of resending raw conversation indefinitely. Preserve provenance for important scores, decisions and tutor notes.	Token growth remains bounded; memory accuracy regression tests pass.

Layer	Requirement	Acceptance evidence
Vendor abstraction	Model/voice providers sit behind internal interfaces so a deprecated model or price change can be swapped without rebuilding product logic.	Provider swap test in staging.
Observability	Trace every request with user/tier/intent/model/cost/latency/retrieval sources/result status. Sample quality safely without exposing unrelated user data.	Per-request trace and cost attribution available to admin.
Rate limits & abuse	Per-user/tier rate limiting, bot/automation detection, account-sharing signals, file/audio caps and anomaly alerts.	Load/abuse test and alert runbook.
Security	Secrets outside prompts/content stores; strict least privilege; signed deep links/actions; tenant isolation; prompt-injection and exfiltration red-team tests.	Security review before production.

40A. Core Data Model and Schemas

These schemas are implementation contracts, not optional documentation. Field names may change, but every concept below must have an explicit schema, ownership and migration strategy before Stage 1 development is estimated.

Entity	Minimum fields / records	Control requirements
Learner profile	user_id; identity references; profession; exam; exam version; exam date; target; availability; language/explanation preferences; AI tier; content entitlements; onboarding status	Consent/source for editable fields; versioning; user-visible edit/delete where applicable.
Event log	event_id; user_id; timestamp; surface; attempt/question/answer; score; video/resource view; submission; plan action; navigation/action result; source/provenance	Append-only event history for analytics/audit; retention policy; no silent rewriting of historical outcomes.

Entity	Minimum fields / records	Control requirements
Learning memory	mastery states; error records; vocabulary; plan state; confidence; tutor notes; AI summaries; source event/source document; created/updated by; edit history	Provenance mandatory for consequential memory; user/tutor correction path; compaction without losing source links.
Credit ledger	ledger_entry_id; user_id; delta; balance-after; provenance (subscription/course/top-up/promo/admin); action charged; purchase/transaction id; expiry; reversal/technical-failure link	Immutable accounting trail; idempotent charges; atomic reversal; no charge on failed technical attempts.
Entitlements	content package; profession scope; AI tier; effective-from/to; purchase channel; grandfathering flag; status	Single authoritative entitlement service used by retrieval, deep links, paywalls and billing.
Knowledge-source metadata	source_id; authority; exam/version; profession; subtest/skill; entitlement; document/video/page/timestamp; approval status; version; checksum; canary/security tags	Supports hybrid retrieval, versioning, leakage controls, rollback and source attribution.

40B. Operational Reliability and Incident Response

Availability target/SLO: TO VERIFY after architecture and hosting are selected. The system must nevertheless have a documented incident process before paid launch.

- Monitor uptime, latency, error rate, model/vendor failures, credit-ledger failures and knowledge-release regressions by surface.
- Define severity levels, on-call/owner, acknowledgement target and customer-communication channel. Numeric targets: TO VERIFY.
- For material incidents, provide an in-product/status communication, preserve credit balances, restore credits for failed chargeable actions, and record the incident timeline.
- Require rollback/recovery runbooks for model/provider, prompt/configuration, knowledge release, payment/credit ledger and voice service.

- Run a post-incident review for critical incidents and feed corrective actions into the risk/evaluation registers.

41. Evaluation and Quality Assurance

Evaluation set	Requirement	Launch gate
Golden questions	Create approved question sets per profession and subtest, including normal questions, ambiguous questions, locked-content attempts, official-vs-methodology conflicts and navigation requests.	Must pass before each prompt/model/content release; threshold TO VERIFY.
Retrieval quality	Measure source recall, precision, reranker quality, citation/source-location accuracy and entitlement filtering.	No critical entitlement leak; numeric targets TO VERIFY.
Hallucination	Track unsupported factual claims, invented locations, invented course rules and false official claims.	Acceptance threshold TO VERIFY; zero tolerance for critical fabricated entitlements/payment actions.
Writing calibration	Compare AI criterion feedback and any score estimate with submissions that have known trustworthy outcomes.	No single numeric band published until calibration is approved.
Speaking calibration	Compare role-play evaluation with human/known outcome data, including linguistic and clinical communication dimensions.	Numeric score/band claims gated until approved.
Arabic set	Maintain a separate Egyptian-Arabic + MSA + Arabic-English code-switching evaluation set.	Pass/fail criteria established during the 2-week spike.
Regression suite	Run before any prompt, model, retrieval, chunking or source update that can affect answers.	Automated regression report attached to release.
Human review	Sample live answers and escalated low-confidence cases. Review rate and reviewer ownership: TO VERIFY.	Documented escalation and correction loop.
Adversarial tests	Prompt injection, system-prompt extraction, Rule Book reconstruction, account sharing, abusive rate patterns and jailbreaks.	Red-team checklist passed before launch.

42. Content Ingestion and Knowledge Operations

	Critical path The AI cannot be “trained on everything” until every source asset is inventoried, processed, tagged, approved and versioned. Content operations is a first-class workstream, not a one-time upload.
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Step	Required work	Owner	Output
1. Asset register	Inventory every Rule Book, PDF, Tutor Book asset, Reading/Listening material, recall, session, workshop, correction session, transcript, video/audio file and platform help article. Record profession/subtest/version/access owner.	Content Ops + Dr Hesham	Master source-of-truth register; counts/hours = TO INVENTORY.
2. Transcription / extraction	Generate or import transcripts; preserve page, slide, audio and video timestamps.	Content Ops / tooling	Searchable raw text with location metadata.
3. Chapter & semantic tagging	Split by lesson/rule/concept; assign profession, exam, skill, difficulty, entitlement and authority tags.	Content Ops + AI engineer	Retrieval-ready structured corpus.
4. Rule extraction	Extract candidate rules, examples, exceptions, FAQs and common mistakes from teaching content.	AI-assisted + human review	Draft structured Rule Book objects.
5. Pedagogical approval	Dr Hesham / named subject owner approves, edits or rejects extracted rules before they become authoritative.	Pedagogy owner	Approved versioned knowledge.
6. Security marking	Apply entitlement labels and optional canary/watermark markers to sensitive proprietary sources.	Security + Content Ops	Protected source set.
7. Evaluation	Run retrieval/golden-question tests and content-diff regression.	AI QA	Release report.
8. Publish / rollback	Publish version with change log; support rollback to prior approved version.	AI Ops	Auditable knowledge release.
9. Ongoing cadence	New workshops/recalls/rule changes enter the same pipeline. Define SLA and maintenance cost.	Content Ops	Continuously current brain; cadence/cost TO VERIFY.

43. Arabic and Voice Feasibility Spike

Run a two-week spike before committing the full Stage 3 voice/Ultimate promise. The goal is to validate production quality and cost, not to prove that a demo can speak.

Test	What to measure	Pass/fail threshold	Fallback if failed
Egyptian Arabic ASR	Recognition of natural Egyptian Arabic medical/exam-prep speech.	TO VERIFY before spike.	Text-first Arabic; push-to-talk with transcript confirmation.

Test	What to measure	Pass/fail threshold	Fallback if failed
Arabic-English code-switching	Mid-sentence English clinical/exam terms inside Arabic conversation.	TO VERIFY.	Text confirmation / language-lock mode.
Latency	Turn-taking delay under realistic mobile networks.	TO VERIFY.	Push-to-talk or non-live voice messages instead of full duplex.
Pronunciation feedback	Whether feedback is educationally reliable across target accents.	TO VERIFY with tutor review.	Restrict to fluency/intelligibility until validated.
Cost	Real per-minute voice + LLM + any telephony/STT/TTS costs.	Must fit approved contribution floor.	All voice stays AI-Credit-metered; reduce included credits or change vendor.
Session controls	12-minute maximum live session, idle/silence cutoff, reconnect handling and no billing for failed technical sessions.	Functional test.	Shorter role-play chunks / async mode.

Current ElevenAgents public pricing lists \$0.08 per minute for additional call minutes and states that external LLM and telephony usage is billed separately. This confirms why large unmetered voice allowances were removed from the base tier design.

44. Legal, Privacy, Trademark and App-Store Readiness

Area	Required action	Status before launch
UK GDPR / data protection	Map data categories/purposes/lawful bases; complete DPIA screening and, where high-risk criteria are met, a DPIA before processing; define data-subject rights and human-review routes.	OPEN - legal/privacy review required.
Voice	Separate notice/consent where required; define whether raw recordings are stored, how long, who can access them and how deletion works.	OPEN.
Sub-processors	List model, voice, hosting, analytics, support and payment vendors; execute appropriate DPAs and document retention/training terms.	OPEN.
Rule Book confidentiality	Contractual no-training/approved-retention position for proprietary content; technical exfiltration controls and audit trail.	OPEN.
Data residency	Future Gulf B2B customers may require regional residency/contractual controls. Treat as B2B-stage requirement, not Stage 1 blocker unless a customer requires it.	DEFER unless contract requires.
OET name/materials	Confirm permitted descriptive/nominative use of OET, disclaimers, ownership of any exam-like content, and whether any official materials are licensed for the intended use.	TO VERIFY with trademark/IP counsel / official terms.

Area	Required action	Status before launch
Jana / Sami	Treat as working persona names only until trademark/domain/app-store/social-handle searches are completed in target regions. A common personal name may be harder to protect as the master mark.	OPEN - clearance required.
Terms of Service	Add AI limitations, educational-only score estimates, academic integrity, acceptable use, account sharing, credit billing/refunds, voice processing and content-upload rights.	OPEN.
Google Play AI policy	Provide in-app reporting/flagging for problematic AI-generated content and maintain safeguards/moderation.	REQUIRED for Play launch.
Apple App Review	Review AI, payments, external-link/steering and account/privacy flows against the storefronts where the app will be distributed.	OPEN; exact storefront terms TO VERIFY.
UK subscription-contract compliance	Design Stage 1 billing/UX for clear pre-contract key terms, auto-renewal reminders, simple cancellation, cooling-off/refund handling and post-trial/renewal rights. The DMCCA subscription-contract regime is expected to commence around January 2027 per the supplied v3 brief; exact commencement and application to the contracting entity are TO VERIFY with counsel.	STAGE 1 DESIGN REQUIREMENT; exact commencement/entity scope TO VERIFY.

45. Risk Register

Risk	Likelihood	Impact	Owner	Primary mitigation
Voice cost overrun	Medium	High	Product + Finance	AI-Credit metering, live cost dashboard, 12-min session cap, silence cutoff, vendor routing.
Rule Book leakage / extraction	Medium	Critical	Security + AI	Entitlement filters, verbatim/volume caps, anomaly detection, canaries, prompt-injection red team.
Writing/Speaking grading inaccuracy	Medium	High	Pedagogy + AI QA	Calibration dataset, uncertainty, no official-score claim, human review samples.

Risk	Likelihood	Impact	Owner	Primary mitigation
Model deprecation / outage	Medium	High	AI Engineering	Vendor abstraction, fallback models, regression suite.
Vendor price change	High	Medium/High	Finance + AI Ops	Per-feature cost attribution, routing, credit pricing, margin alerts.
Store policy / commission change	Medium	High	Product + Legal	Channel-specific economics; review rules before each release; web fallback where lawful.
Arabic/code-switch voice quality	Medium	High	AI + Pedagogy	Two-week spike and fallback modes.
Content ingestion slippage	High	High	Content Ops	Asset register, staged profession rollout, approval SLA.
Key-person dependency	High	High	Founder + Content Ops	Named approvers, versioned rules, documented decision logs and delegation.
Trademark / regulator challenge	Low/Medium	High	Legal	Clearance, disclaimers, IP review, licensed/original content only.
Privacy/data breach	Low/Medium	Critical	Security + Privacy	DPIA/security review, minimisation, encryption, access logs, retention/deletion controls.
Account sharing / abuse	Medium	Medium	Product + Security	Rate limits, device/account signals, fair use and anomaly detection.
Poor free-to-paid conversion	Medium	High	Growth	Contextual paywalls, measurable wow moment, pricing tests, funnel analytics.
Human-support cost creep	Medium	Medium/High	Operations	Defined escalation quotas, self-service, support cost/user dashboard.

Risk	Likelihood	Impact	Owner	Primary mitigation
Development budget materially below scope	High	Critical	Founder + Developer	Before build: developer states exactly which stages/features the ~US\$4,000 quote covers. If it does not cover Stage 1 capability set, cut Stage 1 further or approve additional budget; do not start an unfinishable scope.
UK subscription-law retrofit	Medium/High	High	Legal + Product	Design renewal reminders, simple cancellation, cooling-off/refund handling and key-term disclosure in Stage 1; counsel confirms commencement and contracting entity before launch.
Clinical-boundary misuse	Medium	High	Product + Safety/Pedagogy	Educational-only communication boundary, refusal/redirect for patient-specific clinical decisions, role-play disclaimer and QA/red-team tests.

Risk	Likelihood	Impact	Owner	Primary mitigation
Learner-distress mishandling	Low/Medium	High	Support + Product	Supportive-response standard, no pass-probability/personal-worth framing, human escalation path and safety runbook.

46. Explicit Non-Goals for the First 12 Months

These capabilities remain in the master vision and traceability inventory, but they are not allowed to block OET monetisation or expand Stage 1 scope.

- Full B2B multi-tenancy, white-label tenant administration and enterprise contract features.
- Full IELTS, PTE and TOEFL production packs before OET quality/economics are proven.
- Dedicated Windows and macOS native apps if the web/PWA experience already covers desktop adequately.
- Gamification systems that do not directly improve learning outcomes or conversion.
- Anonymous cohort benchmarking until sample size, privacy and interpretation are reliable.
- SSO, external API, LMS integrations and institution-grade audit exports.
- Fully proactive autonomous coaching before memory, notifications and false-positive safeguards are stable.
- Claims of official exam scoring, guaranteed pass probability or regulator acceptance.
- Unlimited voice or unlimited expensive assessment usage without metering/fair-use controls.
- Patient-specific clinical diagnosis, treatment, prescribing or decision support. The product may teach language and communication around clinical scenarios, but it is not a clinical decision-support system.
- Counselling, psychotherapy or claims about a learner's likelihood of passing. The assistant may be supportive and route distress to human/safety support, but it does not replace professional mental-health care or predict pass probability.

47. Support and Human-Cost Model

Human tutor escalation, support tickets and content corrections are costs, not free features. They must appear below AI cost in the contribution model.

Cost bucket	Measure	Formula	Status
Customer support	Tickets / active user + handling minutes	Total loaded support cost ÷ monthly active users	TO VERIFY from operations data.

Cost bucket	Measure	Formula	Status
Tutor escalation	Escalations / paid user + tutor minutes	Loaded tutor cost for AI escalations ÷ paid active users	TO VERIFY; define tier quotas before advertising.
Content correction	Hours spent reviewing AI/content flags	Monthly content QA cost ÷ paid users	TO VERIFY.
Refund/chargeback	Refunded/charged-back net revenue	Actual refunds + dispute fees ÷ gross billings	TO VERIFY from processor history.
Total contribution	Net revenue - AI - support - tutor - correction - refund provision	Per tier / channel / region	Required dashboard metric.

48. Success Criteria and Target Register

The metrics below must have numeric owners before paid scale. Where the current project has no verified baseline, v3.0 intentionally says TO VERIFY rather than inventing a target.

Metric	Definition	Target	Owner
Free → paid conversion	Registered Free users who become paying AI users	TO VERIFY after beta baseline	Growth
30-day paid retention	Paid users still active/paid at day 30	TO VERIFY	Growth / Product
Monthly voluntary churn	Cancellations excluding failed payments	TO VERIFY; model 2/3/4-month lifetimes	Finance / Growth
Involuntary churn recovery	Failed payments recovered through dunning	TO VERIFY	Finance
ARPU / net ARPU	Gross and net revenue per paid user	TO VERIFY by region/channel	Finance
Net contribution margin	After tax/store/payment, AI, support and refund provision	Minimum floor TO VERIFY; AI-only guardrail ≥75% of net revenue	Finance / Product
LTV:CAC	Cohort contribution LTV divided by acquisition cost	Interim cold-paid guardrail ≥3:1 using current 3-month blended LTV; max CAC ≈ £9.85. Replace with live cohort target before scaling.	Growth / Finance
Support deflection	Eligible support intents solved without human ticket	TO VERIFY	Operations
Thumbs-down rate	Negative feedback / rated AI responses	TO VERIFY	AI QA
Hallucination / unsupported claim rate	Unsupported factual/official/course claims in sampled eval	TO VERIFY; critical entitlement/payment fabrication = zero tolerance	AI QA

Metric	Definition	Target	Owner
Retrieval quality	Recall/precision/citation correctness on golden set	TO VERIFY	AI Engineering
Voice cost / min & completion	Actual full cost and successful role-play completion	TO VERIFY in two-week spike	Product / Finance

49. Final Coverage Verdict - Is Further Evaluation Needed?

Answer No further broad feature brainstorming is required before development. Version 3.0 now covers the learner experience, OET knowledge, monetisation, memory, tutoring, voice, navigation/actions, security, privacy, data schemas, technical architecture, evaluation, content operations, healthcare boundaries, app-store/payment economics, consumer-subscription readiness, owned-audience launch economics, operational reliability, future exams and B2B. Further work is validation/sign-off, not discovery of missing chatbot features.		
Remaining gate	Why it cannot be “finished on paper”	Required result
1. Complete content inventory	Exact video hours, Rule Book/PDF counts and processing effort are not yet enumerated.	Signed asset register + ingestion owner/effort.
2. Arabic/voice 2-week spike	ASR/code-switching quality, latency and real per-minute cost are empirical.	Pass/fail report + final voice-to-credit conversion.
3. Live cost beta	Real token mix, cache hit rate, file/audio usage, voice cost, redemption mix and channel fees determine true margin and the cost of one AI Credit.	One well-instrumented Stage 1 beta: cohort size TO VERIFY; start with one OET profession selected after content readiness review; run for at least 2 weeks; log per-request model/tokens/retrieval/cache/latency/cost, credit redemption, voice minutes, conversion, support events and failed charges. Exit only when target cost per AI Credit, tier ceilings and top-up floors can be approved.

Remaining gate	Why it cannot be “finished on paper”	Required result
1. Complete content inventory	Exact video hours, Rule Book/PDF counts and processing effort are not yet enumerated.	Signed asset register + ingestion owner/effort.
4. Writing/Speaking calibration	A numeric score cannot be trusted without known-outcome data.	Approved calibration report; otherwise criterion feedback only.
5. Legal/privacy/trademark/app-store review	Trademark rights, processor/store terms, privacy duties and UK subscription-contract obligations depend on contracts, regions, commencement dates and exact data flows.	Counsel/privacy/app-store sign-off; contracting entity confirmed; DMCCA commencement TO VERIFY; updated T&Cs/privacy/subscription/cancellation/refund notices.
6. Developer scope/time/budget sign-off	The current vision is materially larger than the referenced ~US\$4,000 development quote; scope must be mapped before work starts.	Developer coverage statement against Stage 1 plus owner + estimate + phase + dependencies. If quote does not reach Stage 1, cut scope further or approve budget before build.
7. Regional price validation	PPP discounts and local payment fees can change contribution materially.	A/B or cohort evidence and approved regional price table.

Scope-control rule: after these gates, new ideas go through change control. They are not evidence that “the chatbot spec was incomplete”. The default should be to ship, measure and iterate against the existing traceability matrix.

Appendix A - Implementation Traceability Matrix

This is the authoritative implementation planning view of the 184-item inventory. “Estimate” is intentionally TO VERIFY until the developer supplies a scoped estimate. Phase/dependency values are planning classifications, not development promises.

Identity & onboarding

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-001	Anonymous demo mode	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-002	Registered Free mode	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-003	Profession selection	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-004	Exam/version selection	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-005	Exam date and target	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-006	Country/regulator goal	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-007	Previous result capture	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-008	Screenshot score import	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-009	Study availability	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-010	Language/code-switch preference	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-011	Teaching-style preference	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema
F-012	Multiple historical exam journeys	☐ Accept ☐ Defer ☐ Reject	Product + Frontend/Backend	TO VERIFY - developer	Stage 1	Auth + learner profile schema

Knowledge & grounding

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-013	All approved Rule Books	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-014	All relevant session transcripts	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-015	Writing workshops	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-016	Speaking workshops	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-017	Correction sessions	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-018	Reading/Listening materials	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-019	Tutor Book	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-020	Dictionaries/common-word lists	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-021	Recalls/practice banks	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-022	Video timestamp index	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-023	Platform navigation map	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-024	Support/FAQ knowledge	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-025	Official current exam sources	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-026	Source hierarchy	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-027	Content versioning	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-028	Teacher override	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources
F-029	Conflict detection	☐ Accept ☐ Defer ☐ Reject	AI/RAG + Content Ops	TO VERIFY - developer	Stage 1-2	Asset register + entitlements + approved sources

Planning & memory

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-030	Initial diagnostic	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-031	Daily plan	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-032	Weekly plan	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-033	30/60/90-day plans	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-034	14/7/3-day plans	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-035	Exam-eve plan	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-036	Missed-day replanning	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-037	Shift-worker planning	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-038	Travel-aware replanning	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-039	What should I do now?	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-040	Next-best-action engine	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-041	Conversation memory	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-042	Learning memory	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-043	Journey memory	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-044	Error DNA	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-045	Learning Fingerprint	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 3	Learner profile + event/memory store
F-046	Spaced reinforcement	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store
F-047	Memory controls/export/delete	☐ Accept ☐ Defer ☐ Reject	AI + Backend	TO VERIFY - developer	Stage 2	Learner profile + event/memory store

Tutoring

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-048	Quick answer	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-049	Detailed tutor	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-050	Socratic tutor	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-051	Examiner mode	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-052	Coach mode	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-053	Dr Hesham mode	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-054	Arabic explanation	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-055	English-only mode	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-056	Adaptive drill generator	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-057	Micro-lessons	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 4-5	Approved pedagogy + retrieval + evaluation
F-058	Grammar tutor	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-059	Vocabulary brain	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-060	Writing guided mode	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-061	Writing hint mode	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-062	Writing correction	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-063	Writing compare/rewrite	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-064	Writing scoring/criteria feedback	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-065	Speaking voice role play	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 3	Approved pedagogy + retrieval + evaluation
F-066	Speaking difficult-patient modes	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-067	Speaking pronunciation/fluency analysis	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-068	Reading Part A/B/C coach	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-069	Listening Part A/B/C coach	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-070	Confidence-vs-accuracy analysis	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation
F-071	Why did my score change?	☐ Accept ☐ Defer ☐ Reject	AI + Pedagogy	TO VERIFY - developer	Stage 2	Approved pedagogy + retrieval + evaluation

Exam & analytics

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-072	Full mock mode	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-073	Practice vs Exam mode separation	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-074	Computer-based rehearsal	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-075	Test-day rehearsal	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-076	Final 24-hours mode	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-077	Result-day assistant	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-078	Resit recovery plan	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-079	Readiness score	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-080	Mastery map	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-081	Trend analytics	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-082	Timing analytics	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-083	Personal bests	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-084	Anonymous cohort benchmarking	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models
F-085	Weekly progress report	☐ Accept ☐ Defer ☐ Reject	AI + Data	TO VERIFY - developer	Stage 2-3	Attempt data + scoring/evaluation models

Multimodal & context

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-086	Image/screenshot understanding	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-087	PDF understanding	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-088	Audio analysis	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-089	Handwritten note understanding	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-090	Score report extraction	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-091	Current-page awareness	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-092	Current-question awareness	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-093	Current-video awareness	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-094	Timestamp-aware help	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-095	Writing selection awareness	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-096	Speaking-session context	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs
F-097	Cross-device conversation continuity	☐ Accept ☐ Defer ☐ Reject	AI + Frontend	TO VERIFY - developer	Stage 2-3	Upload pipeline + page/context APIs

Actions & navigation

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-098	Deep-link resource open	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1	Platform route map + permission-safe action APIs
F-099	Open exact timestamp	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-100	Start recommended practice	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-101	Continue last activity	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-102	Save note	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-103	Save vocabulary	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-104	Add to plan	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-105	Set reminder	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-106	Open workshop	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-107	Create support request	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-108	Tutor handoff summary	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-109	Report wrong answer	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-110	Show allowances/credits	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs
F-111	Contextual upgrade checkout	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1	Platform route map + permission-safe action APIs
F-112	Resume chat after purchase	☐ Accept ☐ Defer ☐ Reject	Frontend + Backend	TO VERIFY - developer	Stage 1-2	Platform route map + permission-safe action APIs

Proactive & engagement

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-113	Exam countdown	☐ Accept ☐ Defer ☐ Reject	Product + Backend	TO VERIFY - developer	Stage 3	Reliable memory + notification preferences

Admin/tutor

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-123	Tutor pre-session brief	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-124	Tutor post-session notes	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-125	Admin aggregate AI queries	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-126	Quality dashboard	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-127	Hallucination/low-confidence queue	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-128	Content-gap dashboard	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-129	Teaching-gap dashboard	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-130	Rule approval/version/rollback	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-131	Content Studio extraction proposals	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-132	Revenue/cost dashboard	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-133	Entitlement management	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log
F-134	Promotional AI Credits	☐ Accept ☐ Defer ☐ Reject	Product + Admin/Backend	TO VERIFY - developer	Stage 2-3	Admin roles + analytics + audit log

Commercial

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-135	Free message cap	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-136	Plus tier	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-137	Pro tier	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-138	Ultimate tier	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-139	AI Credits	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-140	Top-up packs	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-141	Usage counter	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-142	Contextual paywall	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-143	Course × AI entitlement matrix	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-144	Standalone AI subscription	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-145	Course AI add-on	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-146	Monthly billing	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-147	Optional annual billing	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-148	Upgrade/downgrade/cancel	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-149	Fair-use/rate limits	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-150	Regional currency display	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry
F-151	Cost ceiling alerts	☐ Accept ☐ Defer ☐ Reject	Product + Finance/Payments	TO VERIFY - developer	Stage 1	Pricing + payments + entitlement service + cost telemetry

Trust & platform

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-152	Official-vs-methodology separation	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-153	Anti-hallucination behavior	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 1	Security/privacy design + auditability

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-154	Entitlement-safe retrieval	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 1	Security/privacy design + auditability
F-155	Academic integrity controls	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-156	Sensitive upload warning	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-157	Privacy controls	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 1	Security/privacy design + auditability
F-158	Data export/delete	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-159	Audit logs	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-160	Accessibility - WCAG 2.2 AA across learner-facing web, web-app and native-app surfaces	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Design system + accessibility audit + platform-specific implementation
F-161	Arabic RTL support	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-162	iOS	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-163	Android	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-164	Windows	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-165	macOS	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-166	Website	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability
F-167	Web app	☐ Accept ☐ Defer ☐ Reject	Security + AI + Product	TO VERIFY - developer	Stage 2	Security/privacy design + auditability

Expansion & B2B

ID	Requirement	Decision	Owner	Estimate (days)	Phase	Dependencies
F-168	IELTS pack	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-169	PTE pack	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-170	TOEFL pack	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-171	Future exam packs	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-172	Exam-version engine	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-173	Cross-exam skill transfer	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-174	B2B multi-tenancy	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-175	White-label persona	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-176	Institution knowledge upload	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-177	Institution roles	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-178	Seat/usage budgets	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-179	Institution analytics	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-180	SSO	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-181	API/webhooks	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-182	LMS integration	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-183	Custom retention	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture
F-184	Enterprise audit/support	☐ Accept ☐ Defer ☐ Reject	Platform/Enterprise	TO VERIFY - developer	Stage 4-5	Stable OET core + versioned knowledge-pack architecture

Appendix B - Stage Plan with Dates, Budget and Success Gate

Stage	Scope	Start	Target ship	Headcount	Engineering budget	Primary success gate	Dependencies
Stage 0 - Validation	Content inventory; economics baseline; legal/privacy/subscription-law review; data-schema draft; live-beta instrumentation plan; 2-week voice/Arabic spike design	Immediately after v3 sign-off	TO VERIFY	TO VERIFY	TO VERIFY - current total quote must be mapped to scope	All launch blockers have owners; core schemas drafted; beta/voice instrumentation ready; source inventory started; developer quote mapped to stage coverage.	Developer + founder + content/legal owners

Stage	Scope	Start	Target ship	Headcount	Engineering budget	Primary success gate	Dependencies
Stage 1 - OET Core	~15 monetisable OET core capabilities plus cost/credit instrumentation, clear subscription terms, renewal-reminder capability, simple cancellation, cooling-off/refund handling, and incident/kill-switch controls	TO VERIFY	TO VERIFY	TO VERIFY	TO VERIFY	Paid user can onboard, ask grounded eligible OET questions, navigate, get plan, use existing writing assessment, hit contextual paywall; credit ledger/cost dashboard works; required subscription/cancellation controls are build-ready.	Stage 0 decisions + one-profession approved corpus
Stage 2 - Learning Moat	Full approved OET content, Error DNA, adaptive plan, reading/listening/writing tutors, video/timestamps, AI Credits	TO VERIFY	TO VERIFY	TO VERIFY	TO VERIFY	Quality regression + retrieval tests pass; user learning journey persists correctly	Stage 1 telemetry + content pipeline
Stage 3 - Voice & Ultimate	Speaking voice, fingerprint, proactive mentor, multimodal/tutor collaboration	After 2-week voice spike passes	TO VERIFY	TO VERIFY	TO VERIFY	Voice quality/cost gates pass and Ultimate unit economics meet floor	Stage 2 + voice feasibility
Stage 4 - Multi-Exam	IELTS/PTE/TOEFL packs and version engine	TO VERIFY	TO VERIFY	TO VERIFY	TO VERIFY	New exam pack can be added without core rebuild	Stable OET architecture/evaluation
Stage 5 - B2B	Multi-tenancy, white label, institution analytics, enterprise integrations	TO VERIFY	TO VERIFY	TO VERIFY	TO VERIFY	Tenant isolation and first contract requirements validated	Proven B2C economics + enterprise demand

Known commercial constraint: the founder has referenced a US\$4,000 development spend. The developer must explicitly state which stages/features that amount covers; v3.0 does not assume that the entire 184-item vision can be delivered inside that quote.

Stage-budget decision rule: before development starts, the developer must state which stages the quoted budget covers. If the quoted coverage does not reach the approved Stage 1 capability set, Stage 1 is cut further or additional budget is approved before coding begins; the project must not start a scope that cannot be finished.

Appendix C - Verified External Facts and TO VERIFY Register

Verified public references used to correct v1.0 economics/compliance assumptions (checked 30 August 2026).

ElevenLabs - ElevenAgents Pricing: \$0.08/min additional call minutes; LLM/telephony external-provider cost billed separately. [Source](#)

Apple - Choosing a Membership / Small Business Program: Reduced 15% commission for eligible App Store Small Business Program participants. [Source](#)

Apple - App Review Guidelines (UK): External purchase-link/steering rules are storefront/entitlement specific; do not assume US rules apply everywhere. [Source](#)

Google Play - Service Fees: From 30 Jun 2026, UK/EEA/US auto-renewing subscriptions within first \$1M annual earnings: 10% service fee + 5% billing fee when applicable. [Source](#)

GOV.UK - VAT Rates: UK standard VAT rate is 20% for most goods/services. [Source](#)

Stripe UK Pricing: Standard UK cards 1.5% + 20p; international cards 3.15% + 20p, with possible FX fees. [Source](#)

ICO - DPIA / profiling guidance: DPIA required for certain high-risk/systematic profiling and recommended as best practice for new automated profiling; exact applicability depends on processing and effects. [Source](#)

Google Play - AI-generated content policy: Generative-AI apps must provide in-app user reporting/flagging and comply with safety policies. [Source](#)

TO VERIFY before commercial launch

- Exact Apple/Google programme enrollment and storefront-specific commission for the company account.
- Actual payment processor(s), local rail fees, FX costs, VAT/digital-service tax treatment by customer location and merchant entity.
- Final regional PPP pricing and country eligibility.
- Actual content inventory (files, video/audio hours, transcripts, Rule Books, recalls) and ingestion effort.
- Writing/Speaking calibration thresholds and approved disclaimer wording.
- Arabic/code-switch voice pass thresholds, real cost/minute and voice-to-credit conversion.
- Jana/Sami and any master brand trademark/domain/app-store/social clearance.
- OET trademark/nominative-use/material-licensing position and exact disclaimer wording.
- Developer estimates, owners, headcount, stage dates and budget allocation.
- Numeric targets for conversion, retention, churn, LTV:CAC, support deflection and answer-quality thresholds.
- Target fully loaded cost per AI Credit, non-credit usage reserve per tier, and the minimum net-revenue-per-credit multiple for top-up packs.
- Final included AI Credit wallet by tier after the live cost beta; current 2/8/20 values are provisional until the cost rule passes.

- Exact commencement date and entity-level scope of the UK DMCCA subscription-contract regime; approved renewal/cancellation/cooling-off/refund implementation.
- Current deduplicated owned-audience count and channel split behind the stated 52,000+ reach; warm conversion rates by WhatsApp, Telegram, Facebook and YouTube.
- Live beta cohort size and selected first profession; exit criteria must yield enough data to approve cost/credit, latency, quality and conversion decisions.
- Operational availability/SLO, severity-response targets and named on-call/escalation owners.
- Named human owner and operational escalation path for learner distress/safety situations.

NORTH STAR

The AI should know the learner, know the teaching system, know the platform, and know the next best action.